

Neural CFD surrogates earn their keep in the first cell off the wall: a parameter-interpolation baseline and a covariate null for AirfRANS

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Abstract

Machine-learning CFD surrogates are far faster than solvers, but their gains are reported only against other learned models. Kernel interpolation over seven scalars from the AirfRANS case name—no network, no flow-field learning—gives $8.4\times$ lower volume-pressure error than a matched-budget Transolver on an area-uniform 128^2 raster, and **no advantage on any channel** under the dataset’s own node measure, which places 64% of its mesh inside 0.05 chord of the wall against 1.2% of the raster’s area. Re-weighting the identical predictions takes the pressure ratio to **0.44** $\times$ and the streamwise-velocity ratio from $6.2\times$ against the interpolator to $86\times$; the ranking is a property of the weighting, which no published AirfRANS comparison states. The raster’s own round-trip error at those nodes exceeds the surrogate’s by $413\times$: the protocol cannot adjudicate near-wall skill at all. What it does resolve is where each method’s error lives: outward from the wall the interpolation-to-Transolver error ratio on u falls monotonically from $906\times$ to $0.31\times$, while 71.9% of Transolver’s own error lies beyond 0.5 chord. A surrogate buys the first cell and the closure. The benchmark’s primary metric cannot see that: a train-fit regression on the case name ranks official lift at 0.9821 and drag at 0.9318, above all four published lift entries—decisively for only two—and above every drag entry, three of four intervals spanning zero. Nor does the **reynolds** split test Reynolds generalisation: no test case has an in-range inlet velocity, yet a nondimensional interpolator degrades 3% where a dimensional one degrades $5.2\text{--}10.1\times$.

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1. Introduction

Classical CFD resolves the governing PDEs by iterating a discretised system until its residuals fall below a tolerance. It is accurate and trusted, but slow: meshing and a converged steady-RANS solve over a new geometry can take minutes to hours. Neural surrogates promise the opposite trade—millisecond inference—and a vigorous literature now maps geometry plus boundary conditions directly to flow fields with neural operators and attention models.

That promise is assessed almost entirely against other learned models. AirfRANS [1], the standard 2-D external-aerodynamics benchmark, has accumulated a leaderboard of graph networks, point-cloud operators and physics-attention transformers, each reported against its predecessors. What it has not accumulated is a *non-learned* reference point—and the dataset hands one over for free, because every case is indexed by a design vector printed in its own file name and parsed by its own loader: inlet velocity, angle of attack, and four NACA digits.

Research question. What does learning add, on this benchmark, over interpolating between cases in that parameter space—and do the benchmark’s own metrics measure whatever it adds?

The field-MSE ranking is decided by a weighting nobody states. Gaussian kernel-ridge interpolation over seven scalars derived from the case name, with no network, no geometry encoder and no flow-field learning, scored through the identical `evaluate_cases` protocol on the identical 800/200 split, gives $8.4\times$ lower volume-pressure error than a matched-budget Transolver (`mse_p` 75.0 against 628.5) when both arms are scored on the *area-uniform* 128^2 raster this benchmark’s field tables are reported on. Re-weight the *identical* predictions on the *identical* cells by the number of native AirfRANS cloud nodes each cell contains, changing nothing else, and that $8.4\times$ becomes **$0.44\times$** — $2.3\times$ in Transolver’s favour. Every other channel moves with it: `mse_u` from $6.2\times$ against the interpolator to $86\times$ against it, `mse_v` from $3.0\times$ in its favour to $15\times$ against it. Over u, v, p in standardised units the interpolator is $1.99\times$ better area-uniform and $8.30\times$ worse node-weighted. Neither measure is obviously the right one, and that is the point: the raster is what the field tables report, while AirfRANS is a boundary-layer mesh that places 64.4% of every case’s nodes inside $0.05c$ of the wall, a region holding

1.25% of the raster’s area—a factor of 51.5, and 109.5 inside $0.02c$. **No published AirfRANS field comparison states which weighting it uses** (Table 3). Two readings are not in dispute. On the four-channel standardised objective the surrogates are actually trained on, Transolver leads under *both* weightings—by $6.4\times$ area-uniform and by more node-weighted—because ν_t dominates and interpolation cannot represent the closure at all. And the force columns are not volume averages, so no cell weighting reaches them: on lift and drag integrated from the rasterised ground-truth field through the same grid integrator the interpolator is $4.9\times/3.8\times$ **closer** (C_l 1.18% against 5.81%; C_d 2.39% against 8.99%, at 128^2). That pair is a *field-fidelity proxy*, not agreement with AirfRANS’s official force labels: the same integrator carries an $\approx 11\times$ absolute drag bias on *exact* fields and cannot represent the viscous component at all (subsection 4.4).

The deployed protocol cannot adjudicate the boundary layer for either method. Re-weighting fixes the measure, not the representation: both arms are still scored on $r128$ fields, and a method that is right at sub-grid scales earns no credit under either weighting. Resampling the *rasterised ground truth* back at the native cloud nodes measures what the representation costs before any model is involved. That round-trip error exceeds Transolver’s own per-node error by **413** $\times$ pooled over the scored crop and by **495** $\times$ inside $0.005c$ —the band holding 43% of the dataset’s nodes. An $r128$ field metric on AirfRANS is a measurement of far-field skill, for the surrogate as much as for the interpolator, and we read our own field tables that way throughout.

Learning earns its keep in the first cell off the wall. What the two arms do separate, sharply and monotonically, is *where* their errors live. Band by band outward from the wall, the ratio of interpolation error to Transolver error on u runs **906** $\times$ / $74\times$ / $30\times$ / $5.3\times$ / $0.57\times$ / $0.56\times$ / **0.31** $\times$ —monotone across all seven bands, a span of 2905 $\times$, no inversions. Transolver is nearly three orders of magnitude more accurate inside the first half-percent of a chord and a factor of three *less* accurate beyond $0.5c$, where 71.9% of its own squared u error sits against the interpolator’s 3.6%. The interpolator’s side of that boundary was already sharp: **92% of its u error and 90% of its v error lie within 0.02 chord of the wall**—0.5% of cells, thinner than one grid spacing—while **98.7% of the domain**, everything beyond 0.05 chord, is reproduced at $R^2 \geq 0.9996$ with no learning at all (subsection 4.1). These are 128^2 figures. A frozen ladder to 256^2 and 512^2 leaves the domain fraction, the R^2 floor and the outer-region error unmoved, but eases the near-wall shares to 88%/84% because the near-wall error itself falls 45%; our pre-registered rule reads that easing as PARTIAL and we report it as such. Whatever a

learned surrogate buys on AirfRANS, it buys inside the first grid cell and in the turbulence closure—the one channel whose verdict neither weighting disturbs ($14.4\times$ and $9.2\times$ in Transolver’s favour).

The benchmark’s force metrics do not measure that difference. AirfRANS declares the Spearman rank correlation of the force coefficients its primary metric. Ordinary least squares on the case name, fit on the official 800 train cases and scored on the 200 test cases, ranks the official lift at **0.9821** [0.9737, 0.9866] and the official drag at **0.9318** [0.8981, 0.9531]. Under the pre-registered rule—the published mean below the null’s point estimate and outside the null’s 95% interval—all four AirfRANS-paper lift entries fall below that null (0.913, 0.938, 0.965, 0.967; Transolver’s 0.9978 clears). Measured instead against each model’s *own* published seed spread the separation is decisive only twice: 3.8 and 1.9 seed standard deviations for MLP and PointNet, but 1.6 and 0.8 for GraphSAGE and Graph U-Net. We report both readings and rest nothing on the last two. **Drag is where the result is not in doubt:** every AirfRANS-paper entry falls below the null, three of the four published intervals spanning zero (-0.117 , -0.022 , -0.303 , -0.138 , at published seed standard deviations of 0.256, 0.097, 0.124, 0.258). Transolver reports no drag column at all, stating in its Appendix B.1 that the deep models fail at drag. The reading was fixed in the script’s docstring before any number existed, the in-sample optimism of such a fit is 0.0008–0.0035, and the two-parameter version of the null is *not* sufficient—(U, α) alone gives 0.9344 on lift, which one published entry straddles. The result needs the digits, and the digits are fair, because they are the geometry the surrogate is given as an input channel (subsection 4.3).

One of the extrapolation splits does not extrapolate. On the **reynolds** split, 0 of 200 test cases have an inlet velocity inside the training range—outside on both sides—yet a *nondimensional* parameter interpolator degrades by 3% on **mse_u** and not at all on **mse_p**, while the same interpolator on dimensional fields degrades $5.2\times$ and $10.1\times$. The split’s entire difficulty is absorbed by dimensional analysis. The **aoa** split, by contrast, bites exactly where it should: C_l relative error rises from 1.18% to 23.1% (subsection 4.2).

What a surrogate-side physics residual can and cannot do. A deployed surrogate cannot afford the solver that produced its labels, so any self-check it runs must monitor a cheaper, different operator. We audit that substitution and state the verdict in three verbs: the monitored residual can be *ranked* on (AUROC 0.871 on field error, 0.952 on worst-decile drag error—though on field error a physics-free ensemble score matches it within noise), can *certify* only to a floor-limited width (a valid split-conformal band at 0.90 coverage,

6.5–7.9× the drag error it bounds), and cannot be *descended*, because the ground truth itself fails the check (subsection 4.10). We report this because it is measured and because it bounds what self-auditing can deliver, not because it is the headline. The full theorem and evidence for the third verdict are deferred to a companion paper.

The gap this fills. The claim is narrow and we state it as such. Across AirfRANS [1], its Transolver lineage [2, 3], the subsequent architecture surveys [4, 5] and the aerodynamic benchmarks that followed [6, 7], we find *no published parameter-space interpolation baseline for AirfRANS field metrics and no covariate null for its force-rank metric*. The nearest occupant is DriveAerNet++, which does report an AutoML tabular baseline on its 26 design parameters [7]—so the broader claim that nobody publishes parameter-space baselines is false, and we do not make it. What survives is that theirs is an R^2 -on-magnitude comparison rather than a rank one, that the same group’s follow-on architecture benchmark drops it, and that no paper in this line audits *published* force-rank numbers against such a null. The benchmark’s own authors, meanwhile, already located the failure mode without quantifying it: they conclude (their page 8) that “the models have difficulties to predict the wall shear stresses as the velocity values at the closest nodes from the geometry are often largely overestimated”, which “particularly affects the accuracy of the drag coefficient” [1]. This paper supplies the measurement—the same observation from the opposite direction, that near the wall is the *only* place learning helps.

1.1. Contributions

The findings above are the contributions, and the paragraphs that state them carry their numbers; we do not restate them here. What each adds beyond the headline number is:

1. **The missing baseline, the measure that decides it, and the boundary it draws (headline, subsection 4.1).** The aggregate head-to-head is not a stable quantity: on the pre-registered node-count estimator the interpolator wins nothing, and across five discretisations of the node measure `mse_p` ranges over $[0.43, 1.61]$ against 8.39 area-uniform, with no construction leaving the interpolator an aggregate win. What is stable is the decomposition. A wall-distance decomposition localises the entire remaining gap on *both* arms (≥ 0.9965 on the interpolator even with each case’s own band mean removed; a monotone $906\times \rightarrow 0.31\times$ band ratio between the arms), five adversarial

controls—name overlap, near-duplicates, a permuted-parameter negative control, silent case drops, and the solid-fill choice—leave it standing, and 100 training cases of interpolation reach the `mse_p` a 7.35M-parameter Transolver trained on 800 reaches *on the area-uniform measure*. The decision rule was registered before any number existed and returned MIXED; the decomposition is the finding.

2. **The benchmark’s primary metric is a covariate property of the case index (subsection 4.3).** In-sample optimism for such a fit is 0.0008–0.0035, and the (U, α) null alone (0.9344) is *not* sufficient—the digits are, and they are fair, being the geometry the surrogate receives as input. We make no leakage accusation: the claim is that a force-rank number is uninterpretable without this baseline.
3. **The reynolds split does not test Reynolds generalisation (subsection 4.2).** The difficulty is dimensional scaling, not physics; the `aoa` split does bite, and on lift specifically.
4. **What a surrogate-side physics residual can and cannot do (subsection 4.10).** The universal form of the negative is about **iterate selection**—the residual-selected iterate is strictly worse than the error-optimal one in 24/24 cases—and we explicitly do *not* claim descent always harms. The boundary is the operator, not the problem class: where the monitored residual is the *solver’s own*, correction works [8].
5. **A deployable certificate, and an honest account of what it is worth.** A distribution-free split-conformal layer attains target coverage (0.902 ± 0.008 at 0.90), including under distribution shift, and costs a median 1.13ms against a 3822ms solve. As a detector the residual reaches AUROC 0.871 on worst-decile field error, but a *physics-free* ensemble σ does at least as well (0.894) and a paired bootstrap cannot separate them; it is the *fusion* that reaches 0.905 and recovers $\approx 91\%$ of the oracle. The one place physics wins outright is the engineering quantity, $|\Delta C_D|$ (AUROC 0.952). The monotone-residual acceptance gate admits a step on 99.8% of deployed cases and lowers true error on 89.3%—but an *ungated* fixed half-step does better on accuracy alone, so what the gate buys is the certificate, not the accuracy (subsection 4.7, subsection 4.8).
6. **A reproducible, CPU-first open-source package (neuroforge-cfd)** with a frozen I/O contract, interchangeable backbones, an AirFRANS loader, the ablation/certificate harness and the figures, so every claim runs on a laptop and every certificate is reproducible from a committed script.

We are explicit throughout about what is **measured** versus **assumed**, make no out-of-distribution coverage *guarantee*, and make a competitiveness claim only for the Transolver backbone our trust layer runs on (subsection 4.4), not for the grid backbone.

2. Related Work

Neural operators and physics-attention surrogates. Fourier Neural Operators learn a resolution-agnostic mapping between function spaces by parameterising a global convolution in the spectral domain [9]; Geo-FNO [10] extends this to irregular geometries. DoMINO [11] is a decomposable multi-scale point-cloud operator for external aerodynamics, and Transolver [2] replaces quadratic attention with attention over learnable *physics slices*, giving linear cost and strong accuracy; Transolver++ [3] scales it to million-scale meshes. We use Transolver as our matched-budget baseline and as the backbone the trust layer runs on (subsection 4.4). “SOTA” here means its standing as a published state-of-the-art architecture at the time of our experiments (ICML 2024); an independent 2026 benchmark still places Transolver(++) among the strongest aerodynamic surrogates [4], while newer architectures have since advanced the AirfRANS accuracy frontier [5]. None of these carries self-assessment.

Learned solver-correctors and the “fixer” premise. Several lines of work use the discretised residual or a coarse-solution error as a *correction objective*: learned PDE solvers with convergence guarantees [12], deep-equilibrium operators for steady PDEs [13], iterative refiners [14], learned residual-error correctors [15], training-free correction on the PDE residual at each rollout step [16], and PINNs [17], which embed the residual in the training loss. Our contribution to this line is to test the premise head-on and find it does not hold in the deployment-realistic regime: descending $J = \frac{1}{2}\|R_h\|^2$ from the exact ground truth drives the field error up on 200/200 cases (subsection 4.10). The prior successes and this negative are consistent, and what separates them is the *operator*, not the problem class. Residual-driven correction works where the monitored residual is the solver’s own, hence (near-)exact at the true solution—including on steady CFD, where a Newton–Krylov correction lowers the residual by seven orders of magnitude *while* reducing field and aerodynamic error [8]—so the boundary is demonstrably not steady versus transient. It fails where the monitor is instead a surrogate-side operator that a deployed system can afford, inconsistent with the one that produced the labels.

Residual-based, data-free error signals, and conformal UQ. Closest to our trust-signal half is work reading the PDE residual as a reference-free error indicator: Roy et al. [18] trigger a classical solver when a residual-based estimator fires in time-marching; Song et al. [19] align epistemic-UQ bands with localised residual structure; Gopakumar et al. [20] use the residual itself as a conformal nonconformity score, obtaining coverage in *residual* space with the transfer to solution space left open. The classical answer to the same question is that residual magnitude becomes an error estimate only when weighted by an adjoint solution [21], and its neural analogue bounds a PINN’s true error by its residual scaled by a stability constant [22]; both are conditional on the residual vanishing at the exact solution, which our monitor does not do. Mukherjee et al. [23] prove the corresponding convergence statement only under additional compactness assumptions. On the UQ side, deep ensembles [24] and MC-dropout [25] are standard epistemic estimators, and conformal prediction gives distribution-free coverage for operator learning [26], PINNs [27], neural operators [28] and automotive surrogates [29]. We propose no new UQ method; we calibrate on the field the user actually receives—the corrected one—and additionally test, and reject, the residual’s use as a correction objective.

Benchmarks, and what they compare against. AirfRANS [1] provides ~ 1000 incompressible steady-RANS simulations over NACA airfoils with `full`, `scarce`, `reynolds`, and `aoa` splits, and declares the Spearman rank correlation of the force coefficients its primary metric. Its reported entries—and those of the Transolver lineage [2, 3] and subsequent surveys [4, 5]—are compared against one another and against earlier learned models, never against a model of the design parameters that index the cases. The same holds of the aerodynamic benchmarks that followed [6], with the one partial exception already stated: DrivAerNet++’s AutoML tabular baseline on its 26 design parameters [7].

Evaluation and reporting standards in ML for PDEs. A parallel line asks whether the reported gains in this field mean what they appear to mean. McGreivy and Hakim [30] find that 60 of 76 surveyed ML-for-PDE papers compare against a weak baseline, and identify outcome-reporting bias as systematic rather than occasional. Our two evaluation results are instances of exactly those failure modes, made concrete on one benchmark and with a constructive replacement: a baseline that runs in fifteen minutes of CPU time, and a null any author can compute for their own force numbers. We do not claim published AirfRANS results are wrong, and we specifically do not claim label leakage—conditioning a surrogate on its inlet state is necessary,

not suspect. We claim these numbers have been read against a missing reference point.

Simple-baseline audits: the lineage both of our reference points belong to. Neither control is a new kind of instrument. In natural-language inference, *partial-input* baselines that discard the premise recover much of the label and revised what those benchmarks were taken to measure [31, 32]. In graph classification, Errica et al. [33] showed by comparison against *structure-agnostic* baselines that on several benchmarks structural information was not yet being exploited—substitute *geometry* for *structure* and that is the sentence we are testing here. In recommendation, Ferrari Dacrema et al. [34] found most reproducible neural methods matched by nearest-neighbour or linear models. The genre has since reached the natural sciences: Ahlmann-Eltze et al. [35] report that no tested deep or foundation model for gene-perturbation effects *yet* outperforms deliberately simple linear baselines. Feng et al. [36] supply the standing caveat on the instrument and we honour its direction: a cheap baseline that succeeds shows a benchmark is cheatable, while a cheap baseline that fails licenses nothing about how hard the benchmark is. One structural difference separates our controls from the partial-input line and is worth stating precisely. A partial-input baseline consumes a subset of what the model consumes; the design vector we read off the case name is, for the most part, not an input to any AirfRANS surrogate—it is the low-dimensional code that *generated* the geometry, published in the file name. AirfRANS is a hybrid case and we flag it as such: *U* and *Re* are literally input channels, while the NACA digits enter the surrogate only through the *sdf* they generate. Framed generally, the question is how much usable information a benchmark’s own published metadata carries about its headline label [37]; we answer it in the benchmark’s own currency—volume MSE and Spearman rank—rather than as an information-theoretic quantity.

The estimator is decades old, and we say so first. Gaussian kernel ridge regression over a handful of design variables is Kriging in all but name, and fitting response surfaces over design variables has been standard practice in aerodynamic design since long before this benchmark existed [38]. We therefore claim no novelty for the estimator; an aerodynamicist should read [subsection 4.1](#) as response-surface methodology used as a benchmark control, not as a new method. What is new is the measurement—what the omission of that estimator from this benchmark’s reporting conceals, and where the surviving gap lives. That is also the cleanest statement of how this work extends the closest prior art in our own subfield: McGreivy and Hakim [30] showed that the field compares against weak *numerical* solvers; we show that

it does not compare against a trivial *statistical* baseline at all, and we measure how much that conceals. The protocol we recommend is likewise not an invention but an import. Data-driven weather forecasting already mandates trivial controls of exactly this kind, reporting persistence and climatology alongside every model in its headline scorecard [39]; external-aerodynamics benchmarking does not.

2.1. Positioning

This work sits in the evaluation line rather than the architecture line. Its primary contribution is a pair of reference points for AirfRANS—a non-learned field baseline and a case-name force-rank null—and the boundary they draw: what a learned surrogate adds is localised to the first grid cell off the wall and the turbulence closure, and the benchmark’s declared primary metric cannot resolve it. Running the field baseline also exposes a property of the benchmark’s field metric itself, which is the result we lead with: the aggregate ranking of a 7.35M-parameter surrogate against a seven-scalar interpolator is decided by whether the volume average weights cells by area or by native node count—a convention no published AirfRANS comparison states—and the raster both are scored on cannot resolve near-wall skill for either of them. The secondary contribution is an audit of the self-check a deployed surrogate could run instead, whose organising question is not which features a system has but whether the residual it drives is the *same discrete operator* that produced its reference data. Where it is, residual-driven correction works and is already demonstrated [8]. Where an affordable surrogate-side proxy stands in for it— ~ 1.1 ms per case against a solver-grade operator per step—the truth itself fails the check. The trust-layer components this paper retains are individually well developed in the recent literature [20, 29, 18, 19, 23] and we claim no priority over them; what is new is the measurement that constrains all of them.

3. Method

The framework wraps a one-shot backbone f_θ in a verify–estimate–correct loop: the backbone predicts, a physics residual and an uncertainty estimate are computed, a split-conformal layer turns them into a calibrated trust map, and a learned corrector applies a damped update under a monotone-residual acceptance test. This section defines only the parts later sections measure. Implementation details, the module map and the synthetic zero-download data generator are in the released package and its `docs/REPRODUCE.md`; no quantitative claim in this paper uses synthetic data.

3.1. Geometry encoding and the monitored operator

A **FlowCase** is encoded into a fixed channel-first stack $x \in \mathbb{R}^{7 \times H \times W}$ in the frozen **INPUT_CHANNELS** order (**sdf**, **mask**, x , y , u_{in} , v_{in} , $\log \text{Re}$); the backbone outputs **OUTPUT_CHANNELS** (u, v, p, ν_t) , where p is the **kinematic** pressure p/ρ . Fixing this contract is what makes the backbones interchangeable and the trust layer backbone-robust.

The verifier evaluates the steady, incompressible, 2-D RANS equations in primitive form on the **physical (denormalised)** fields, with effective viscosity $\nu_{\text{eff}} = \nu + \nu_t$, pointwise:

$$\text{continuity: } r_c = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}, \quad (1)$$

$$x\text{-momentum: } r_x = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + \frac{\partial p}{\partial x} - \nu_{\text{eff}} \nabla^2 u, \quad (2)$$

$$y\text{-momentum: } r_y = u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + \frac{\partial p}{\partial y} - \nu_{\text{eff}} \nabla^2 v. \quad (3)$$

Because p is kinematic, no explicit density appears. **The turbulence term is live:** over the fluid cells of the 200-case AirfRANS test split the eddy viscosity averages $\nu_t = 8.2 \times 10^{-5}$ against a laminar $\nu = 1.56 \times 10^{-5}$, so $\nu_t \approx 5.2\nu$ supplies $\approx 84\%$ of ν_{eff} and the monitored operator is a genuine RANS residual. The channel is learned, backbone-dependently: against ν_t 's own fluid-masked variance, $R^2 = 0.996$ on the grid backbone and 0.67–0.70 on MeshGraphNet (`scripts/check_nut_learned.py`). What the operator omits is the spatially varying term $\nabla \nu_t \cdot \nabla u$. Derivatives use central finite differences with one-sided stencils at the borders; residuals are zeroed inside the solid. A boundary-condition violation map r_{bc} adds a **no-slip** term penalising velocity magnitude in the thin fluid band adjacent to the wall (weighted by $\exp(-|\text{sdf}|/\ell)$, $\ell \approx 3$ cells) and a **far-field** term penalising deviation from (u_∞, v_∞) on the outer border ring.

Two implementation facts that bear on later claims. (i) The monitor discards its own strongest no-slip cells, because a central stencil on the fluid ring adjacent to the solid reaches into the $u = v = 0$ discontinuity; the masking is necessary but removes a mean 46% (range 35–51%) of the squared no-slip penalty—an unlogged spot check over 8 cases, which we report as such—precisely where the proximity weight is largest. *(ii) There are two Laplacian stencils in this pipeline.* The numpy monitor whose numbers we report uses the compact three-point form; the differentiable twin used to build the corrector's residual *input* channels uses a repeated first difference, which is wider and annihilates the period-2 mode the compact form sees. This is a

live alternative explanation for the null result of the residual-input ablation (subsection 4.4) and we do not claim to have excluded it; where it could have mattered most, in the descent experiment, we check both operators explicitly and both fall together in 24/24 cases.

3.2. Trust map, conformal calibration, and the correction operators

Raw trust map. The combined PDE-residual magnitude $\rho_{\text{res}} = \sqrt{r_c^2 + r_x^2 + r_y^2 + r_{bc}^2}$ and a per-cell predictive uncertainty σ are each mapped to $[0, 1]$ (using an absolute physical reference scale $s = U_\infty^2/L + U_\infty/L$ when available, else a robust 95th-percentile normalisation with an absolute floor) and fused,

$$e = \text{clip}(w_r \hat{r} + w_u \hat{\sigma}, 0, 1), \quad T = 1 - e, \quad (4)$$

with $w_r = 0.6$, $w_u = 0.4$. Uncertainty σ is the channel-mean standard deviation from a deep ensemble or MC-dropout.

Case-level versus per-cell scope. Our quantitative trust result is **case-level**: the field-mean residual ranks which whole predictions to distrust (per-case Spearman 0.611, subsection 4.4). The *per-cell* spatial rank correlation with error *within* a field is weaker— 0.22 ± 0.06 on the dropout-FNO at $n=15$, and 0.166 ± 0.155 on the deployed ensemble-mean field at $n=200$ —but scale-dependent: patch-pooling the map to 16×16 cells roughly doubles it (0.323 ± 0.232 , $+0.157$, improving 93% of cases; Gaussian smoothing at $\sigma=4$ cells gives $+0.135$), and the pooled patches flag top-decile-error *regions* at AUROC 0.68–0.72; dynamic-pressure normalisation does not help ($+0.008$) (`results/control/percell_localization.json`). The residual is therefore a *case-level* ranker and a *moderate region-level* locator, not a per-cell error rank; per-cell claims are left to the conformal band, which uses the calibrated σ .

Split-conformal calibration. Raw σ is uncalibrated, so a threshold on it carries no guarantee. On a held-out calibration set we compute per-channel non-conformity scores $s = |\hat{y} - y|/\sigma$ and take the finite-sample-corrected $(1 - \alpha)$ quantile q ; the band $q \cdot \sigma$ then satisfies Proposition 1 (cf. 26).

Proposition 1 (Distribution-free coverage under exchangeability). *If the calibration scores and a test score are exchangeable, then the split-conformal band $q \cdot \sigma$ satisfies $\mathbb{P}(|\hat{y} - y| \leq q \sigma) \geq 1 - \alpha$.*

Corrector and acceptance test. A small residual CNN predicts an additive correction conditioned on the current field, its residual map and the geometry, $\Delta_k = c_\phi(y_k, R(y_k), x)$. A candidate $y_{k+1} = y_k + s \Delta_k$ is admitted

only if it does not increase the residual norm $N(y) = \sqrt{r_c^2 + r_x^2 + r_y^2}$, i.e. $N(y_k + s \Delta_k) \leq N(y_k) + \varepsilon$, with s halved up to four times on failure; by construction $N(y_0) \geq \dots \geq N(y_K)$. The test is satisfiable by the identity, so a corrector that cannot reduce the residual is simply rejected—but measured, backtracking finds an admissible damped step on 99.8% of deployed cases (subsection 4.8). The deployed corrector is instead a Deep-Equilibrium model [40]: the correction δ is the fixed point of $T_\theta(\delta; c) = \kappa \cdot g_\theta([\delta, c])$ with $c = [\hat{y}, r(\hat{y}), \text{geom}]$, g_θ spectrally normalised [41] and $\kappa < 1$, trained with Jacobian-Free Backpropagation [42].

Proposition 2 (Banach contraction of the corrector). *With g_θ each-layer 1-Lipschitz and $\kappa < 1$, T_θ is a κ -contraction in δ . By the Banach fixed-point theorem the equilibrium δ^* exists, is unique, and the iteration converges geometrically, $\|\delta_k - \delta^*\| \leq \kappa^k \|\delta_0 - \delta^*\|$.*

Crucially g_θ is trained on **data**—it targets the correction toward truth, with the residual as an input *feature*—rather than by minimising the residual, and at inference δ^* is applied directly, **without** the acceptance test. Two scopes must be kept distinct: the DEQ contraction is in the *correction variable* δ , whereas the *PDE residual of the output field* is a different quantity that can rise even as δ converges and the field error falls. After the loop, a trust-gated **ClassicalFallback** would hand low-trust regions to an external solver; in the present release it is an **interface with a stub backend** that reports what would run without invoking one, and it bears on no quantitative claim.

4. Experiments

All quantitative results are on **real AirfRANS**. Unless noted, the ablations use the pre-registered protocol (`benchmarks/ablation.py`, `docs/EXPERIMENTS.md`) over **3 seeds (0, 1, 2)**, reported as mean $\pm$ std (population std, `ddof=0`; sample std at $n = 3$ widens bars by $\approx 1.22\times$). Metrics follow the AirfRANS community protocol (`evaluate_cases`): per-channel volume MSE—**area-uniform over the fluid cells of the r128 crop** in every table except Table 3, which re-scores the same predictions under the dataset’s own node measure and is where subsection 4.1 shows why the choice matters—surface-pressure MSE on the body, Spearman rank correlation of the force coefficients ρ_{C_l} , ρ_{C_d} , and **residual_error_spearman** (> 0 means a low residual tracks low error). Throughout the tables, ρ_{C_l} and ρ_{C_d} correlate forces integrated from the predicted against the ground-truth field through our grid surface-integrator—a *field-to-field self-consistency* valid for

relative arm comparisons, not agreement with AirfRANS’s official force labels (subsection 4.4). The ν_t channel is omitted from the tables because its raw MSE is not comparable to the others; measured against its own fluid-masked variance it is learned, backbone-dependently. With $n = 3$ we report **per-seed sign-consistency** plus effect-size magnitude rather than p-values. An effect called “robust” is sign-consistent across all 3 seeds with a large effect size and, where stated, non-overlapping per-seed distributions.

The grid backbone is an FNO trained on AirfRANS full (800 train / 200 test, 80 epochs) in four arms: `backbone`, `backbone (no physics loss)`, `backbone + local corrector`, and `backbone + DEQ corrector`. The certificate runs use a dropout-enabled FNO (width 48, 4 layers, 20 modes, dropout 0.05) with a DEQ corrector ($\kappa = 0.9$, damping 0.5), trained 40 + 15 epochs at resolution 128. The v2 phase (subsection 4.4) re-runs the trust layer and DEQ corrector on a SOTA Transolver backbone under the same protocol.

4.1. Where learning earns its keep, and the measure that decides the ranking

AirfRANS indexes every case by a short design vector that the dataset’s own file name prints and its own loader parses: the fields of `airFoil2D_SST_U_alpha_digits` are the inlet velocity, the angle of attack and the four NACA digits, and `Simulation.reset()` reads the first two directly off the name. Before asking how well a neural operator predicts these flows, we ask what that index alone already determines.

The baseline. From the name alone we build seven analytic scalars—inlet velocity U , geometric incidence α , NACA maximum thickness $t_{\max}$, thin-airfoil effective incidence $\alpha - \alpha_{0L}$, and the camber line at $x/c \in \{0.25, 0.5, 0.75\}$ —and interpolate the *nondimensional* field $(\hat{u}, \hat{v}, \hat{p}, \hat{\nu}_t) = ((u - U \cos \alpha)/U, (v - U \sin \alpha)/U, p/U^2, \nu_t/(Uc))$ across training cases with Gaussian kernel ridge regression, redimensionalising with the test case’s own (U, α) . There is no network, no geometry encoder and no flow-field learning; fitting is one 800×800 Cholesky solve and prediction is one GEMM. The estimator is deliberately old—this is response-surface methodology, and Gaussian kernel ridge over design variables is Kriging in all but name [38]—so the claim we make is for the measurement, not for the method. Hyperparameters come from 5-fold cross-validation *inside* the train split on a scalar registered before any number existed, applied to test once. Scoring is byte-identical to the surrogates’: the same cached rasterised pairs at 128^2 , the same `evaluate_cases`, the same mask, surface sampler and force integrator, with `sdf` and `mask` rebuilt from `case.geometry` exactly as the Transolver adapter does (`scripts/parameter_interpolation_baseline.py`,

Table 1: Parameter interpolation against learned surrogates on AirfRANS full (800 train / 200 test, r_{128} , identical `evaluate_cases`, 3 seeds where seeded). The force columns are mean relative error against the force integrated from the rasterised *ground-truth field* through the same grid integrator—a field-to-field consistency measure, not agreement with the official AirfRANS labels—and ρ_{C_l}, ρ_{C_d} are the corresponding self-consistency correlations defined in [section 4](#). That integrator carries an $\approx 11\times$ absolute drag bias even on exact fields and cannot see viscous drag ([subsection 4.4](#)), so these four columns rank arms against one another; they do not measure force accuracy. The last block is the floor ladder and the negative control; “permuted” predicts each test case from another case’s parameters under a fixed derangement, everything else identical. The interpolator’s 0 parameters are not free: it carries the 800 training fields (210 MB) resident at inference. Transolver’s force-rank seed standard deviations are 0.0002 (ρ_{C_l}) and 0.0012 (ρ_{C_d}), kept out of the cells for width. **Bold marks the better arm under the area-uniform cell weighting these volume columns use.** `mse_v` and `mse_p` carry no bold because their ordering reverses under the dataset’s own node measure on these same predictions ([Table 3](#)); the surface, force and `mse_u` columns do not reverse.

model	n_params	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_l}	ρ_{C_d}	C_l err	C_d err
Transolver (baseline)	7.35M	0.120±0.005	0.088±0.013	628.5±29	9110±500	0.9992	0.9963	5.81%	8.99%
our grid backbone (Table 9)	—	3.479	0.385	2444.8	548823	0.987	0.895	—	—
our grid backbone + DEQ	—	3.457	0.880	3832.7	361681	0.986	0.923	—	—
parameter interpolation (no learning)	0 (210 MB)	0.782	0.0336	75.03	10989	0.9999	0.9991	1.18%	2.39%
floor: 8-NN, inverse distance	0	0.904	0.143	1145.5	108022	—	—	—	—
floor: 1-NN	0	1.415	0.308	2744.0	192123	—	—	—	—
floor: train-mean field	0	11.01	6.577	57980	2792467	—	—	—	—
floor: uniform freestream	0	26.01	17.65	151365	4120337	—	—	—	—
control: permuted parameters	0	22.82	14.36	129622	5939745	0.345	—	365%	367%

`results/interpolation/interp_full.json`). The input asymmetry runs *against* the baseline: the surrogates receive the full 7-channel geometry and freestream encoding per case, the interpolator receives seven scalars.

The *objective* asymmetry runs the other way, and it is large enough to decide the comparison. The interpolator is selected by cross-validation on the area-uniform grid measure our tables report; the surrogate minimises a per-point loss under the dataset’s own node distribution, which places 55.5% of its nodes inside $0.02c$ against 0.51% of the raster’s cells—a factor of 109.5, rising to $312\times$ inside $0.005c$ and inverting to $0.14\times$ beyond $0.5c$ ([Table 3](#) and the paragraph after next). [Table 1](#) puts the baseline against the learned arms under the area-uniform measure, [Table 3](#) re-scores the identical predictions under the node measure, and [Figure A.7](#) plots the grid backbone’s gap to Transolver in the same comparison.

Result, and the weighting it is contingent on. On the area-uniform 128^2 measure the interpolator beats a matched-budget Transolver on cross-flow velocity (`mse_v` 0.0336 against 0.088, $2.6\times$), on volume pressure (`mse_p` 75.0 against 628.5, $8.4\times$), and on both force columns (C_l relative error 1.18% against 5.81%, $4.9\times$ lower; C_d 2.39% against 8.99%, $3.8\times$ lower). **Neither volume ratio survives the dataset’s own node measure, so each is**

quoted here as a pair and should be quoted that way or not at all (Table 3): `mse_p` is 75.0 against 629.6 area-uniform and 6012 against 2646 node-weighted, a ratio of $8.39\times$ in the interpolator’s favour against $0.44\times$; `mse_v` is 0.0336 against 0.0999 area-uniform and 4.26 against 0.282 node-weighted, $2.97\times$ against $0.066\times$. Only the cell weights differ between those two columns—same predictions, same truth, same fluid mask, same 200 cases, same per-case-mean protocol—and before the re-weighting is applied both arms reproduce their published rows to below one part in 10^9 , the Transolver arm exactly (`scripts/measure_asymmetry.py` gates **G1** and **G2**; `results/interpolation/measure_asymmetry.json`).

The force columns carry no cell weighting to change, so they are untouched, but they need their name said out loud, because it is not the name a reader supplies: they measure agreement with the force integrated from the rasterised *ground-truth field* through the same grid integrator, so the $4.9\times$ and $3.8\times$ are field-to-field consistency ratios and not $4.9\times$ and $3.8\times$ better lift and drag. On the official OpenFOAM labels the same integrator is $\approx 11\times$ biased on absolute drag even when fed exact fields and cannot see the viscous component at all (subsection 4.4), and against those labels the interpolator’s own drag rank (0.8389) is indistinguishable from the exact truth field’s (0.8394). The interpolator loses under *both* weightings on the streamwise channel (`mse_u` 0.782 against 0.120, $6.5\times$; 30.7 against 0.358, $86\times$ node-weighted), marginally on surface pressure (10989 against 9110, $1.21\times$), and on the turbulence variable (`mse_nut` 8.8×10^{-8} against 5.7×10^{-9} , $15\times$; on Table 3’s re-scored rows, $14.4\times$ area-uniform and $9.2\times$ node-weighted—the only channel whose margin narrows rather than reverses). One tension between those readings we record without resolving: the interpolator is $1.21\times$ worse on surface-pressure MSE yet $4.9\times$ better on the lift column, which is an integral of that same surface pressure, so the two statistics must weight the body differently and we have not decomposed which cells drive each. Because `run_baselines.py` trains on per-channel *standardised* MSE over all four output channels, we report that metric too, so the comparison cannot be dismissed as scoring a model on something it was never trained for (Table 2). Both of its means must be quoted together, and both are area-uniform: **over the three channels the tables report the interpolator is $1.9\times$ better, and over all four channels of Transolver’s actual loss Transolver is $6.4\times$ better**, because ν_t dominates both models’ standardised error and the interpolator cannot represent the closure at all. Recomputed from Table 3’s own rows against the same $\text{Var}_{\text{train}}$ divisors—which are common to both arms and so cancel in every ratio—the area-uniform pair is $1.99\times$ and $6.1\times$, differing from Table 2 only in the provenance of the Transolver row, and the node-weighted pair reverses to $8.30\times$ and $8.85\times$ in Transolver’s

favour. We do not describe this as beating Transolver on its own objective under any weighting. What the two tables show is the *shape* of each method’s error: Transolver’s accuracy is concentrated—13× better on u than on p in standardised terms—and it is the only method here that gets ν_t right; area-uniform, the interpolator is uniform across u , v and p and is blind to the closure.

“Weight by node count” is not a single number, and none of its readings rescues the aggregate. The node measure admits more than one discretisation, so we re-derive the same ratio under five of them from committed numbers, with no new inference (the `sensitivity` stage of `scripts/measure_asymmetry.py`). On `mse_p` they span $[0.43, 1.61]$ against 8.39 area-uniform, and the endpoints are not equally credible. The two faithful cell-level estimators—sample the grid error field at each node, then average over nodes, pooled or per case—give 0.431 and 0.440; the coarse but internally consistent band-level construction gives 1.10; the largest value, 1.61, comes from the two constructions that pair true node masses with cell MSEs drawn from a different spatial set, which we report because they are what a reader would compute by hand from [Table 5](#) and because they are the least favourable to our own conclusion. **No construction leaves the interpolator an aggregate win:** the construction that reaches 1.61 on p is 551× against it on u and 24× on v , and across all five node constructions `mse_u` runs 86–551× against it and `mse_v` 6.3–24×. Two properties of the re-weighting run the same way. A $0.0234c$ cell cannot sit inside a $0.005c$ band, so the grid assigns only 0.098 of the node weight to the innermost band against its true 0.432, moving mass *out of* the band where Transolver leads by 906×; and 13.4% of in-crop nodes bin to cells the geometry mask calls solid and are dropped, those being the nodes nearest the wall. The true node measure is therefore more adverse to the interpolator than 0.440, not less.

The boundary is one grid cell wide, and that is the contribution. Decomposing the interpolator’s squared error by signed distance to the wall (`scripts/interpolation_band_control.py`, `results/interpolation/interp_band_control_full.json`) locates the entire remaining gap. **92% of its u error and 90% of its v error lie within $0.02c$ of the wall—a band holding 0.5% of cells, thinner than the $0.0234c$ grid spacing—while 98.7% of the domain lies beyond $0.05c$, where it reproduces every channel at $R^2 \geq 0.9996$ ([Table 5](#)).** We report the strict per-case-centred statistic alongside the pooled one, because U spans 31–93 m/s and a pooled far-field R^2 partly rewards knowing the

Table 2: The same runs in per-channel standardised MSE—each channel divided by its own train-set variance (u 341.9, v 52.45, p 135590, ν_t 1.320×10^{-6}), which is the quantity the surrogates are trained on, with $F_OUT = 4$. The last two columns are the two means, and neither should be quoted without the other. Every cell here is **area-uniform**; the divisors are common to both arms, so the row ordering is the ordering of Table 1 and reverses with it under the node measure. Applying these same divisors to Table 3’s rows gives $1.99\times$ and $6.1\times$ area-uniform and $8.30\times$ and $8.85\times$ node-weighted, both in Transolver’s favour in the second case.

model	std. mse_u	std. mse_v	std. mse_p	std. mse_nut	mean u, v, p	mean u, v, p, ν_t
Transolver	3.51×10^{-4}	1.68×10^{-3}	4.64×10^{-3}	4.34×10^{-3}	2.22×10^{-3}	2.75×10^{-3}
parameter interpolation	2.29×10^{-3}	6.41×10^{-4}	5.53×10^{-4}	6.70×10^{-2}	1.16×10^{-3}	1.76×10^{-2}
our grid backbone	1.02×10^{-2}	7.34×10^{-3}	1.80×10^{-2}	—	1.18×10^{-2}	—

freestream: subtracting each case’s own band mean first leaves $R^2 \geq 0.9965$ on every channel beyond $0.05c$, and moves only the sub-cell wall band ($0.840 \rightarrow 0.754$ on u). This cuts both ways and both cuts matter. It defeats the obvious objection that the win is on easy cells: the far field is 80% of the domain and carries 3.6% of the u error and 2.2% of the p error. And it says precisely what a learned surrogate buys on this benchmark: the first cell off the wall, the streamwise channel that carries the boundary-layer and wake deficit, and the turbulence closure. Those are exactly the three channels the surrogate wins—**mse_u**, surface pressure sampled 1.5 cells off the wall, and ν_t —so the decomposition and the head-to-head agree on where the advantage is. One scope note travels with the wall-band number: the concentration is 0.924 on u and 0.898 on v but only 0.535 on p , so “the advantage is confined to 0.5% of cells” is a statement about the velocity channels, not about pressure.

The same decomposition on the surrogate, and the boundary becomes a crossover. A decomposition of one arm shows where that arm fails; it does not show that the other arm succeeds there. We therefore run the identical band accumulator on the deployed Transolver—same cases, same rasterisation, same fluid mask, reproducing Table 5 over 95 entries at a maximum relative deviation of 6.9×10^{-8} (gate **G3**)—and report the per-band ratio of the two (Table 4). On u it runs **$906\times / 74\times / 30\times / 5.3\times / 0.57\times / 0.56\times / 0.31\times$** from the wall outward: **monotone across all seven bands, a span of $2905\times$, no inversions**, with the crossover between $0.02\text{--}0.05c$ and $0.05\text{--}0.15c$. Transolver is nearly three orders of magnitude more accurate than parameter interpolation inside $0.005c$ and a factor of three *less* accurate beyond $0.5c$. The two arms’ error distributions are mirror images: 92.4% of the interpolator’s u squared error lies inside $0.02c$, while 71.9% of Transolver’s lies beyond $0.5c$ (and 62.8% of its p error), in the 80% of cells where the in-

Table 3: **The ranking is the weighting.** The same 200 test cases, the same rasterisation and fluid mask, the same predictions and the same per-case-mean protocol; *only the cell weights differ*. “Area-uniform” gives every fluid cell equal weight, which is what Table 1 reports. “Node-weighted” gives each fluid cell the number of native AirfRANS cloud nodes that bin to it, which is the measure the dataset itself carries. Bold marks the better arm within each row; the last two rows are Transolver over interpolation, so a value above 1 means the interpolator is better by that factor and is bolded. Before re-weighting, the interpolator arm reproduces `interp_full.json` to below one part in 10^9 and the Transolver arm reproduces `v2_results.json backbone_per_seed` exactly (gates **G1/G2**); Transolver is the mean over the three deployed seeds (per-seed R_p : 8.90/8.66/7.61 area-uniform, 0.418/0.433/0.469 node-weighted). The area-uniform Transolver `mse_p` is 629.6 here against 628.5 in Table 1, which transcribes `results/baselines/table2.csv`; the two differ only in provenance. The re-weighting is conservative—grid binning cannot place node mass inside a sub-cell band, and solid-binned nodes are dropped—so the true node measure is more adverse to the interpolator than the node-weighted row shows. Source: `scripts/measure_asymmetry.py`, `results/interpolation/measure_asymmetry.json`.

cell weight	mse_u		mse_v		mse_p		mse_nut	
	interp	Transolver	interp	Transolver	interp	Transolver	interp	Transolver
area-uniform ($r128$ crop)	0.782	0.127	0.0336	0.0999	75.0	629.6	8.84×10^{-8}	6.15×10^{-9}
node-weighted (same cells)	30.7	0.358	4.26	0.282	6012	2646	4.72×10^{-7}	5.12×10^{-8}
ratio, area-uniform		0.162		2.97		8.39		0.070
ratio, node-weighted		0.0117		0.0663		0.440		0.108

terpolator carries 3.6% of its own. **That is why an aggregate field-MSE table is decided by its weighting:** the two methods are accurate in different places, by three orders of magnitude, and any aggregate is a choice about where to look. The p row makes the point concretely. Transolver is better in the innermost band (0.589), the interpolator wins modestly from 0.005 c to 0.05 c , and then by $32\times$ and $237\times$ in 0.15–0.5 c and beyond 0.5 c : *the area-uniform $8.4\times$ on `mse_p` is a far-field statement about 80% of the cells and 11% of the nodes.*

Two sample-size facts belong with the innermost number rather than being left for a reviewer to find. Pooled over 200 cases the 0–0.005 c band holds 4489 fluid cells (22.4 per case) against 2 595 931 beyond 0.5 c ; the $906\times$ is a paired ratio over those 4489 cells, both arms evaluated on exactly the same cells. What makes it hold up is that it sits on a monotone trend rather than alone, and that 0.005–0.01 c (3824 cells) and 0.01–0.02 c (8143 cells) independently give $74\times$ and $30\times$. It is also the band the resolution ladder already refines: at 512^2 , where it is 3.4 cells across instead of sub-cell, the interpolator’s own near-wall error *falls* 45% while the far field moves 0.6%, so a finer grid would narrow this ratio rather than inflate it.

Neither arm’s near-wall skill is observable at $r128$, and that bounds every number in this section. AirfRANS is a boundary-layer mesh, and the mis-

Table 4: Per-band ratio $r_b = \text{MSE}_{\text{interp}}(b)/\text{MSE}_{\text{Transolver}}(b)$ on the AirfRANS full test split, r_{128} , 200 cases; $r_b > 1$ means Transolver is better in that band by that factor. Transolver is the mean over the three deployed seeds. Band edges are signed distance in chord units, identical to Table 5. “Cell frac.” is the band’s share of the r_{128} crop’s fluid area; “node frac.” its share of the native cloud’s nodes; the last column is Transolver’s own share of squared u error, whose mirror image is the interpolator’s in Table 5. Pooled fluid-cell counts are in the artifact for every band. The accumulator reproduces Table 5 over 95 entries at max relative deviation 6.9×10^{-8} , and $\sum_b n_b \text{MSE}_b$ reconstructs the pooled squared error for every arm, channel and band grid (gates **G3/G4**). Source: `results/interpolation/measure_asymmetry.json, verdicts.D2`.

band (chord)	cell frac.	node frac.	u	v	p	ν_t	SE share u , Transolver
0–0.005 c	0.00138	0.4322	906.2	48.3	0.589	393.1	0.0057
0.005–0.01 c	0.00118	0.0612	74.1	3.30	2.49	65.6	0.0045
0.01–0.02 c	0.00251	0.0619	30.4	1.17	1.58	15.7	0.0094
0.02–0.05 c	0.00742	0.0884	5.33	0.391	1.22	4.44	0.0210
0.05–0.15 c	0.02942	0.1118	0.572	0.113	0.259	3.70	0.0560
0.15–0.5 c	0.15831	0.1311	0.559	0.030	0.031	9.45	0.1847
> 0.5 c	0.79978	0.1134	0.312	0.007	0.004	21.1	0.7188

match between its node distribution and a uniform raster is not marginal. Over the 200 test cases its clouds carry 35.9 M nodes, 95.4% of them inside the $3c \times 3c$ crop the metric scores. 43.2% of those nodes lie inside 0.005 c of the wall, a region holding 0.138% of the crop’s area—a factor of 312. Cumulatively, 55.5% lie inside 0.02 c (0.51% of the area, $109.5 \times$) and 64.4% inside 0.05 c (1.25% , $51.5 \times$); beyond 0.5 c the asymmetry inverts to $0.14 \times$. The spread across cases is negligible—the node fraction inside 0.05 c has minimum 0.6045, median 0.6163 and maximum 0.6238—so this is a property of the mesh generator, not of a few cases, and it does not depend on which signed-distance function defines the bands: the grid `sdf` and the cloud’s own distance column agree on 99.51% of nodes at the 0.02 c threshold and 99.90% at 0.05 c (gate **G6**).

Re-weighting fixes that measure, not the representation both arms are still scored on. To bound the representation, we resample the *rasterised ground truth* at the native nodes and compare its round-trip error against the model’s, which is the error the scoring representation commits before any model is involved. Inside 0.005 c Transolver’s per-node u error is 1.16; the r_{128} raster’s own is **572.8**. Pooled over the crop the factor is **413** $\times$; inside 0.005 c it is **495** $\times$. **The scoring representation destroys, in the region holding 43% of the dataset’s nodes, structure 495 $\times$ larger than the model error it is being used to adjudicate.** Every field number in this section—ours, and the published rows we set them against—is therefore a statement about far-field skill. This is the half of the objection a re-weighting

Table 5: Wall-distance decomposition of the interpolator’s error on AirfRANS full test. R_{pc}^2 subtracts each case’s own band mean before scoring, so it asks whether spatial structure *within* the band was captured beyond case-level scale. “SE share” is the fraction of the channel’s total squared error in that band and is unaffected by the centring choice. The pooled (non-centred) R^2 quoted in the text is higher than R_{pc}^2 at every entry; both are in the artifact.

band (chord)	cell frac.	$R_{\text{pc}}^2 u$	$R_{\text{pc}}^2 v$	$R_{\text{pc}}^2 p$	SE share u	SE share v	SE share p
0–0.02 c	0.005	0.7539	0.9722	0.9966	0.924	0.898	0.535
0.02–0.05 c	0.007	0.9929	0.9990	0.9986	0.018	0.040	0.238
0.05–0.15 c	0.029	0.9991	0.9997	0.9996	0.005	0.030	0.154
0.15–0.5 c	0.158	0.9983	0.9999	0.9999	0.017	0.018	0.052
> 0.5 c	0.800	0.9965	0.9999	1.0000	0.036	0.014	0.022

cannot reach, and it is symmetric: not a concession about the interpolator but a limit on what an $r128$ AirfRANS field metric can say about any method. (We read this from the in-crop block of the artifact; outside the crop the bilinear sampler clamps query points to the domain boundary, so the outer bands of the full-cloud block are clamp-contaminated and we do not quote them.)

Under refinement the localisation holds and the share statistic dilutes: a pre-registered PARTIAL. Those shares are 128^2 numbers, and the obvious attack is that the 0.02 c band is thinner than the 0.0234 c cells it is measured on. We re-ran the whole decomposition at 128^2 , 256^2 and 512^2 on the same 800/200 split with the estimator frozen—the kernel weight matrix is a function of the case *names* alone and hashes bit-identically at every rung—and with every band edge fixed in chord units (`scripts/interpolation_resolution_ladder.py`, `results/interpolation/interp_resolution_ladder.json`; the 128^2 rung reproduces Table 1 at relative error 0.00e+00, which gates the run before any finer number is written). **The pre-registered verdict is PARTIAL** and we report it as such. The rule required the near-wall squared-error share to stay at or above 0.85 on *both* u and v at every rung; the shares go 0.924/0.898 at 128^2 to 0.901/0.855 at 256^2 to **0.879/0.842** at 512^2 , so v misses the floor at the finest rung. That 0.85 was written before any number existed and was not moved, and no clause of the DISSOLVES branch fires.

The share erodes because its numerator shrank, not because the far field got worse—and that reading is a stronger result than the deployed-resolution table could show. Inside 0.005 c , which is 0.15% of the domain, the absolute u error *falls* 45% under refinement ($466.7 \rightarrow 257.7$) while its R^2 rises $0.507 \rightarrow 0.723$; beyond 0.5 c , which is 80% of the domain, the absolute u error moves 0.6% ($0.0356 \rightarrow 0.0354$). A share is a ratio, so when the dominant term

halves and the rest holds, every other band’s share rises without any of them degrading. What survives unqualified is everything the objection asserts. At 512^2 , where the band is 3.4 cells across rather than sub-cell, **72%** of the u error lies inside $0.005c$: the localisation is *tighter* at fine resolution, not looser. The fraction of the domain reproduced without learning is $0.9949/0.9949/0.9950$; every channel still clears $R^2 \geq 0.9996$ beyond $0.05c$ at all three rungs, though we note that the binding entry— p in 0.05 – $0.15c$ —clears it at 0.99961 at every rung, so that claim rests on a fourth-decimal margin at any resolution; and absolute MSE beyond $0.05c$ is flat within 1.5% across a $16\times$ increase in cell count. Two scope notes travel with the ladder. At 512^2 the raster spacing is only $1.28\times$ the source cloud’s far-field nearest-neighbour spacing, so the outer-MSE arm of the rule was pre-declared to be read on the $128 \rightarrow 256$ pair, clear of that limit. And there is no Transolver row at 256^2 or 512^2 : the ladder settles whether the *localisation* is an artifact of the deployed grid, and settles nothing about whether the *advantage* survives at native resolution (section 5).

The ladder also re-scores the force columns, and they turn out to be ill-conditioned in a way a reader should be told about. By 512^2 the interpolator’s C_d relative error rises $2.4\times$ on the median and $2.6\times$ on a 90%-trimmed mean while its *absolute* drag error *falls* ($0.0085 \rightarrow 0.0060$); the untrimmed mean reaches 1.84, but that is one case, because `coefficient_metrics` divides by a rasterised reference drag that collapses and changes sign on that case under refinement ($+0.146$ to -1.7×10^{-4} ; the case is named in the artifact). ρ_{C_l} is unmoved at 0.9998 and ρ_{C_d} slips $0.9991 \rightarrow 0.9811$, most of it recoverable by dropping that one case. With no Transolver row at those rungs, and with the rasterised reference integral itself moving under refinement, the 128^2 force ratios cannot be re-evaluated here—only the interpolator’s own trend, and only with the metric’s ill-conditioning stated.

How we tried to break it. Five checks could have killed the result and did not. *Leakage*: train/test case-name overlap is 0 on all three splits, asserted in the script and verified independently against `manifest.json`. *Near-duplicates*: the minimum nearest-neighbour distance in standardised feature space is 0.189 (median 0.440), and excluding the closest 10% of test cases makes the aggregate *worse*, not better (`mse_p` $75.0 \rightarrow 82.1$), so the headline is not carried by a few near-copies. *Degenerate metric*: predicting each test case from another case’s parameters collapses the result to the freestream floor (last row of Table 1; ρ_{C_l} $0.9999 \rightarrow 0.345$, C_l error $1.18\% \rightarrow 365\%$), so the signal is in the parameters, not in the rasterisation or the mask. *Silent drops*: `coefficient_metrics` swallows failed force integrations, so the script asserts $n = 200/200$ (and $196/196$ on `aoa`) on every scored variant. *Self-serving solid*

fill: filling pressure inside neighbours’ solids by nearest-fluid propagation instead of leaving the Delaunay bridge makes surface pressure $14\times$ worse ($10989 \rightarrow 153174$), so the one channel the surrogate wins is not flattered by our treatment. Run along the same resolution ladder, that control localises its own sensitivity to the coarse grid: the two defensible fills disagree by $13.7\times$ at 128^2 , $5.8\times$ at 256^2 and $1.7\times$ at 512^2 , because at $h = 0.0234c$ the bilinear surface stencil straddles the body while at $h = 0.0059c$ it sits in real fluid. We therefore make no claim about surface pressure and grid resolution—the apparent $1.9\times$ degradation under **nearest** reverses to $4.2\times$ the other way under **nearest_all**—and we read our $r128$ surface-pressure number as what it is, a number about the sampler and the solid fill as much as about the flow. Selection never saw test: the cross-validated ranking inside train and the test ranking agree, with kernel ridge and local linear filling the CV top ten and every k -NN configuration $33\times$ behind.

It needs the shape, and it needs few examples. Restricted to (U, α) the baseline collapses (**mse_p** 9491); adding thickness gives 7507; the full seven features give 75.0. The airfoil digits are worth $126\times$ on **mse_p**, so this is not a freestream-lookup table. Conversely, dropping U from the feature metric while keeping the nondimensional rescaling barely moves anything ($75.0 \rightarrow 93.4$), which shows U entering as pure dimensional scaling—the mechanism behind [subsection 4.2](#). On a train-size ladder with five random subsets per size, 100 **training cases** of parameter interpolation take **mse_p** to 238 ± 65 (worst of five subsets 344) and **mse_v** to 0.0720 ± 0.013 , which on the area-uniform measure is past the 7.35M-parameter Transolver trained on all 800 (628.5 and 0.088). At 50 cases the mean already clears **mse_p** (519) but one subset of five does not (735), so the claim we make is 100, not 50. Convergence in **mse_p** has not saturated at 800. The ladder’s internal result—an eighth of the training data buys the estimator’s full accuracy—is a statement about the interpolator alone and is measure-free; the cross-method comparison inside it is not, and carries [Table 3](#) with it.

The cost asymmetry, and what this does not say. The interpolator has zero parameters but is non-parametric: it carries $800 \times 128^2 \times 4 \text{ float32} = 210 \text{ MB}$ of training fields, all resident at inference, against Transolver’s 7.35M parameters ($\approx 29 \text{ MB}$). That is a real objection to calling it a deployable surrogate. It is not an objection to reporting it as the baseline, which is the only claim we make for it. Nor is this a claim that published AirfRANS results are wrong: it is a claim that they have been read against a baseline nobody ran, and that the baseline is strong enough to change how they read. We also deliberately do *not* place a NeuroForge row in a cross-method comparison: AirfRANS results are reported in mutually incompatible conventions (standardized-field

MSE and official OpenFOAM force labels in the AirfRANS paper, relative- L_2 in the Transolver lineage, physical-unit MSE and self-integrated forces here), so any shared cell would mislead. [Table A.12](#) reproduces the canonical AirfRANS-paper baselines as field context, and [subsection 4.3](#) adds the row that makes it readable.

4.2. The *reynolds* split does not test Reynolds generalisation

The same estimator, re-selected by cross-validation inside each split’s own train set, separates the benchmark’s two extrapolation tasks cleanly. The script proves disjointness rather than asserting it: on `reynolds`, 0 of 200 test cases have an inlet velocity inside the training range— $U \in [46.8, 78.0]$ train against $[31.5, 93.4]$ test, outside on both sides—and on `aoa`, 0 of 196 have an in-range incidence.

Yet the nondimensional interpolator barely notices `reynolds`: `mse_u` degrades by 3% ($0.782 \rightarrow 0.808$) and `mse_p` by -1% ($75.0 \rightarrow 74.1$), and it is *better* than in-distribution on `mse_v` and on surface pressure. The control that explains why is in the same run: the `raw` cell-wise interpolator, identical except that it does not nondimensionalise, degrades $5.2\times$ on `mse_u` ($0.808 \rightarrow 4.194$) and $10.1\times$ on `mse_p` ($74.1 \rightarrow 751.1$) on that split. **The split’s entire difficulty is absorbed by dimensional analysis, not by learning.** Physically this is unsurprising — Re spans $2.0\text{--}6.0 \times 10^6$, all fully turbulent—but the consequence for evaluation is sharp: a “Reynolds OOD” result on this split is not evidence of learned extrapolation unless the method is shown not to be exploiting u/U and p/U^2 .

The `aoa` split does bite, and only where it should. Lift is the quantity that depends on incidence, and the C_l relative error rises from 1.18% to **23.1%**; `mse_u` degrades $2.1\times$ and `mse_p` $1.7\times$, while drag is barely touched ($2.39\% \rightarrow 2.71\%$). That is the expected signature of extrapolating past the stall-approach end of the α range. One caveat is load-bearing: no matched *surrogate* row exists on either split in our results, so these numbers report the interpolator’s own in-distribution-to-ODD degradation and must not be read as “interpolation generalises better than a surrogate”.

4.3. The benchmark’s force metrics do not measure that difference

[subsection 4.1](#) locates the surrogate’s advantage in the first cell off the wall. The benchmark’s declared primary metric cannot see it. Bonnet et al. [1] name the Spearman rank correlation of the force coefficients as “the main metrics to maximize the recovery of the best airfoils”—and that rank is a covariate property of the case index.

We fit ordinary least squares on the case name—inlet velocity, incidence, incidence squared, and the four NACA digits—on the official 800

Table 6: The case-name null against the published AirfRANS `full` field. Nulls are OLS fit on the official 800 train cases and scored on the 200 test cases; intervals are 95% percentile bootstrap over 10^4 case-level resamples. Published rows are from Bonnet et al. [1] Table 3 and Wu et al. [2], quoted with their own seed standard deviations where reported; Transolver reports no drag column.

	ρ_{C_l} vs official	ρ_{C_d} vs official
null: α	0.9335 [0.9114, 0.9483]	0.8001 [0.7249, 0.8563]
null: (U, α)	0.9344 [0.9123, 0.9492]	0.8276 [0.7601, 0.8768]
null: (U, α, α^2)	0.9344 [0.9123, 0.9492]	0.8752 [0.8278, 0.9083]
null: $(U, \alpha, \alpha^2, \text{NACA})$	0.9821 [0.9737, 0.9866]	0.9318 [0.8981, 0.9531]
MLP	0.913 $\pm$ 0.018	−0.117 $\pm$ 0.256
GraphSAGE	0.965 $\pm$ 0.011	−0.303 $\pm$ 0.124
PointNet	0.938 $\pm$ 0.023	−0.022 $\pm$ 0.097
Graph U-Net	0.967 $\pm$ 0.019	−0.138 $\pm$ 0.258
Transolver	0.9978	not reported

train cases, and score Spearman against the official labels on the 200 test cases: the exact protocol the published entries were produced under. The reading was fixed in the script’s docstring before any number existed (BELOW THE NULL iff the published mean is below the null’s point estimate *and* the null’s 95% bootstrap interval excludes it). The null ranks official lift at **0.9821** [0.9737, 0.9866] and official drag at **0.9318** [0.8981, 0.9531] (Table 6; `scripts/covariate_null_trainfit.py`, `results/review/covariate_null_trainfit.json`).

Every baseline in the AirfRANS paper sits below the full null on both coefficients, on the metric that paper declares primary—four of the five published entries once Transolver is included, which clears on lift (0.9978) and reports no drag column at all, stating in its Appendix B.1 that “all the deep models fail in drag coefficient estimation”. The drag column is worse than the gap alone suggests: **three of the four published drag intervals span zero** at one published seed standard deviation (−0.117 $\pm$ 0.256, −0.022 $\pm$ 0.097, −0.138 $\pm$ 0.258; only GraphSAGE’s −0.303 $\pm$ 0.124 excludes it, and on the wrong side). Four qualifications belong with this, and we state them before a reviewer does.

On lift the margin is decisive for two entries, not four. The rule above compares a published mean against the null’s point estimate and its interval, and all four AirfRANS-paper lift entries fall below it on that rule. A reader who compares instead against each model’s own published spread will not see four clean results: the gap is 3.8 published seed standard deviations for MLP (0.913 $\pm$ 0.018) and 1.9 for PointNet (0.938 $\pm$ 0.023), but only 1.6 for

GraphSAGE (0.965 ± 0.011) and 0.8 for Graph U-Net (0.967 ± 0.019). The two intervals are not commensurable in any case: ours is a case-level bootstrap over the 200 test cases, theirs a spread over five training seeds, and neither carries the other’s variance source. The summary we stand behind is therefore that the null’s point estimate exceeds all four lift entries and separates decisively from two of them. None of this qualification applies to drag, where the null clears every entry by a wide margin and three of the four intervals span zero.

The digits are fair. The NACA digits are the geometry the surrogate is handed as an input channel, so a null that uses them is not exploiting information the models lack; it is asking whether the models extract anything from that geometry beyond what a seven-coefficient regression already extracts. Restricting the null to (U, α) —the reading an earlier version of this control used—gives 0.9344 on lift, which PointNet straddles and MLP sits just below at 1.2 seed standard deviations. **The two-parameter null is not sufficient to support this claim; the digits are what make it decisive.**

The fit is honest. The numbers above avoid in-sample optimism by construction, being fit on train and scored on test. A companion run that fits and scores entirely within the 200 test cases bounds the optimism directly: the gap between the in-sample and the 10-fold cross-validated Spearman is 0.0017 on lift and 0.0020 on drag at the full feature set, and 0.0008–0.0035 across all four feature sets (`scripts/covariate_null.py`, `results/review/covariate_null.json`). Covariate regressions of this form carry negligible optimism at this sample size.

This is a statement about the metric, not an accusation. We do not claim label leakage. A drag *rank* metric cannot distinguish a better integrator from a head that has learned the covariate dependence, and conditioning a surrogate on its inlet state is necessary rather than suspect. The defensible claim is the measurement gap: no paper in this line reports the null, so its force-rank numbers are uninterpretable as evidence about integration quality. The claim is also narrower than “nobody publishes parameter-space baselines”, which is false as stated ([subsection 4.1](#)).

4.4. Backbone-agnostic trust and learned correction on a SOTA backbone

We re-run the trust layer and the DEQ corrector on a **SOTA Transolver backbone** (7.35M parameters, 80 epochs, AirfRANS full, 5 seeds; corrector trained 20 epochs with the backbone frozen), the strongest model in our study. Protocol and scoring are identical to [subsection 4.5](#); the sources are `results/v2/v2_results.json` (original 3-seed study) and `results/control/w1_capture.json` (identical recipe, extended to five

Table 7: SOTA Transolver backbone (7.35M params, 80 ep, AirfRANS `full`), alone vs. with the DEQ correction loop. 5 seeds, mean $\pm$ std (population std, `ddof=0`). Lower MSE is better; ρ closer to 1 is better; `resid $\leftrightarrow$ err` $\rho > 0$ supports the trust signal. The C_l/C_d columns are relative error against the ground-truth field’s own integrated forces, not against the official labels; the paragraph after next states what that integrator can and cannot see. Source: `results/control/w1_capture.json` (5-seed extension of `results/v2/v2_results.json`).

metric	mse_u	mse_v	mse_p	C_l rel. err	C_d rel. err	resid $\leftrightarrow$ err ρ
backbone	0.133 $\pm$ 0.017	0.096 $\pm$ 0.006	644 $\pm$ 63	5.7%	7.6%	0.611 $\pm$ 0.054
+ DEQ loop	0.122 $\pm$ 0.015	0.076 $\pm$ 0.006	485 $\pm$ 39	5.5%	6.8%	0.612 $\pm$ 0.046
Δ	−8%	−21%	−25%	−0.2pp	−0.8pp	+0.002

seeds; the extension reproduces the 3-seed corrector deltas and ρ_{C_l} within pre-registered tolerances).

Learned correction works on a strong backbone. The DEQ corrector reduces volume field MSE on **all five seeds**: `mse_u` −8%, `mse_v` −21%, `mse_p` −25%. **Surface-pressure** MSE is the one channel it does *not* improve: essentially flat at 9843 $\rightarrow$ 9899 (+0.6%; per-seed −1/−1/+2/+2/+1%). Internal force-ranking consistency is held at $\rho_{C_l} = 0.999$ and $\rho_{C_d} = 0.996$. These deltas are within-backbone—the same architecture against itself, on one measure—so they are untouched by Table 3. What Table 3 does touch is the cross-method reading: area-uniform, the corrected `mse_p` = 485 is $6.5\times$ the 75.0 a parameter interpolator reaches with no learning, while node-weighted the ordering is the other way round. Under either weighting the −25% is a relative improvement to a learned backbone and not a field-accuracy record; its value lies in the channels where interpolation fails— u , ν_t and the first cell off the wall—and in being reference-free at deployment, which an interpolator carrying 210 MB of training fields is not.

What these force correlations measure, and what they do not. Our ρ_{C_l}/ρ_{C_d} correlate the force integrated from the *predicted* field against the force integrated from the *ground-truth* field through the same grid surface-integrator, not agreement with AirfRANS’s official OpenFOAM labels. Recomputed against those labels (`scripts/recompute_force_vs_official.py`; 200 cases, 3 seeds; `results/control/force_vs_official.json`), the deployed backbone recovers force *ranking*— $\rho_D = 0.84\pm 0.01$, $\rho_L = 0.88$ —but not magnitude: absolute drag is biased ($\approx 11\times$ mean relative error) by the integrator’s 1.5-cell surface-sampling offset, and the same integrator on *perfect* fields also yields $\rho_D = 0.84$, so this number is not limited by the prediction. Nor is it limited only by the measurement: a three-parameter OLS on (U, α, α^2) that never touches a flow field ranks the official drag

at $\rho = 0.874$ —above every integrator arm we tried on ground-truth fields (0.611–0.839) and on predicted fields (0.828–0.845); α alone gives 0.800 (`results/control/drag_covariate_control.json`). The parameter interpolator makes the same point from the other direction: scored through this integrator it ranks the official drag at $\rho = 0.8389$, *indistinguishable from the same integrator applied to the exact ground-truth field* (0.8394), and matching it on lift to four significant figures (0.8758 vs 0.8757). **A drag rank correlation reported on this benchmark is uninterpretable without its covariate baseline**, and the conventional near-field number sits below that baseline.

Where the loss lives, and what is repairable. Rasterising the exact truth to 128^2 , scattering it back to the source mesh nodes and integrating with AirFRANS’s *own force_coefficient* (four pre-registered gates all exactly 0.0) gives $\rho_D = 0.700$ and $\rho_L = 0.9967$ —drag *below* our own 0.839, so the raster, not the quadrature, owns the loss; controlling for the covariates leaves partial $\rho_D = 0.133$ against partial $\rho_L = 0.976$. The mechanism is resolution: 68% of drag is viscous and the wall shear lives in a sublayer at $y^+ \approx 5$, so recovering it needs $N \sim 10^5$ cells where lift, set by chord-scale surface pressure, converges at first order over the same ladder: lift is observable from a raster at deployed resolution and drag is not. Repairing what is repairable (`results/control/force_vs_official_multischeme.json`), a wall-extrapolated pressure scheme cuts the ground-truth-field lift magnitude error from 16% to 4.6% (median), and a *control-volume* far-field momentum balance recovers official lift from ground-truth fields almost exactly ($\rho_L = 1.00$, median error $< 1\%$) and from *predicted* fields at $\rho_L = 0.998 \pm 0.001$ with per-seed median magnitude errors of 3.6–3.9%: engineering-grade lift from the surrogate. Drag decomposes diagnostically—on ground-truth fields the control-volume estimator cuts the median drag error from $3.3\times$ to $0.23\times$, but on predicted fields it reaches only $\approx 2.7\times$ with a degraded ranking, so the residual drag error is the *model’s* far-field accuracy, not the integrator. We therefore report lift as quantitatively recovered and make no claim of accurate absolute drag. Two limits travel with this. $\rho_L = 0.998$ is an easy number twice over: α is an input channel and lift here is close to a function of it, and the control-volume estimator recovers lift from circulation through the outer ring, so it depends on the far field and not on the surface at all. And with the solid-adjacent ring masked at $Re \approx 10^6$, the momentum thickness is far below one cell, so the *viscous* component of drag is not merely inaccurate here—it is **unavailable**; every drag statement in this paper is about the pressure-dominated component, which is also why we never report $C_{d,v}$.

The gain comes from the learned correction, not from feeding it the residual (W1 control). To isolate *why* the corrector helps, we train the same DEQ corrector two ways under identical architecture, data, optimiser, schedule and seeds, differing only in whether the 3 residual input channels carry the real residual (WITH) or are permanently zeroed at both train and inference (NULL). Across 5 seeds the null corrector **matches or slightly exceeds** the deployed one: the per-seed paired difference is negative on all 5 seeds for both `mse_u` and `mse_speed` (mean ≈ -0.005 on a ≈ 0.12 base, $\approx 4\%$; WITH beats NULL on 0/5 seeds; `results/control/w1_capture.json`). The improvement is attributable to the learned correction architecture, not to residual-conditioning—consistent with the residual’s role elsewhere: it tells you *which* predictions are wrong, not *how* to fix them. The deployed system keeps the residual input because it is already computed for the trust layer, so conditioning on it costs nothing extra. One caveat qualifies the null ([section 3](#)): those input channels are built from the wider differentiable stencil, whose larger kernel we have not excluded as an explanation.

The trust signal is backbone-robust. On this SOTA backbone the bare residual–error correlation is already strong (0.611 ± 0.054 over five seeds—per-seed 0.611/0.652/0.613/0.667/0.511, positive on every seed), against ≈ 0.40 for the bare weak backbone ([Table 9](#)); the corrector barely moves it (0.612 ± 0.046). We add a third, architecturally distinct data point: a message-passing GNN (MeshGraphNet, param-matched at 7.35M, 3 seeds, bare backbone), where the bare case-level residual–error Spearman is 0.851 ± 0.058 (`results/mgn/mgn_results.json`). Across a spectral grid Geo-FNO, an attention/point-cloud Transolver and a message-passing GNN the correlation is consistently *positive*, though variable in strength. **The MeshGraphNet arm is not a like-for-like third point:** that model was trained on 16k-point subsamples and evaluated at the full $\approx 180\text{k}$ points, and our own density control (`results/control/mgn_density_control.json`) returns DENSITY-DRIVEN with an MSE ratio of 2.44. The control measures MSE on four cases and carries no rank correlation, so it does not establish that 0.851 is *itself* a density artifact—but the 0.40–0.85 range should be read as 0.40–0.61 among the arms trained and evaluated at matched density. The detector remains positive on all three.

Coverage holds under both uncertainty estimators; only an ensemble is adaptive. Split-conformal calibration of the pressure channel at $\alpha = 0.1$ attains target coverage with *either* estimator: 0.902 ± 0.008 with MC-dropout (3 seeds) and 0.915 with a deep ensemble (single run), both at the 0.90 target. Coverage was never the difficulty—**adaptivity** was. With MC-dropout this

Table 8: Conformal calibration of the pressure channel on the SOTA Transolver backbone ($\alpha = 0.1$, 100/100 cal/test split): MC-dropout vs. a deep ensemble ($M=5$). Both attain target coverage; only the deep ensemble yields an input-adaptive band (finite q , low ECE). MC-dropout: 3 seeds (mean $\pm$ std); deep ensemble: single ensemble training (split-to-split spread over 20 calibration/test re-draws: coverage 0.899 ± 0.009 , $q = 2.26\pm 0.05$; [subsection 4.7](#)). Sources: `results/v2/v2_results.json`, `results/uq_ensemble/uq_results.json`.

σ estimator	coverage (target 0.90)	q	ECE ($\downarrow$)	band
MC-dropout ($p=0.05$)	0.902 ± 0.008	$\approx 1.6 \times 10^9$	≈ 0.31	near-constant
deep ensemble ($M=5$)	0.915	2.35	0.074	adaptive

Table 9: In-distribution AirFRANS full ablation, 3 seeds, mean $\pm$ std (population std). Lower MSE is better; ρ closer to 1 is better; `resid $\leftrightarrow$ err` $\rho > 0$ supports the trust signal. Bold marks the best arm per column among the physics-loss arms; the physics-loss-free arm (starred where best overall) is reported separately because it removes the physics objective entirely. See [Figure 1](#).

arm	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_l}	ρ_{C_d}	<code>resid$\leftrightarrow$err</code> ρ
backbone	3.479 ± 0.105	0.385 ± 0.012	2444.8 ± 120.7	548823 ± 27946	0.9868 ± 0.0005	0.895 ± 0.013	0.397 ± 0.003
backbone (no phys. loss)	1.995 ± 0.042	$0.323\pm 0.008^*$	$1963.5\pm 55.5^*$	1123649 ± 104822	0.9920 ± 0.0002	0.945 ± 0.008	0.605 ± 0.016
backbone + local corr.	3.482 ± 0.058	0.398 ± 0.007	2469.4 ± 71.0	541534 ± 20453	0.9854 ± 0.0017	0.888 ± 0.012	0.389 ± 0.031
backbone + DEQ corr.	3.457 ± 0.811	0.880 ± 0.064	3832.7 ± 331.3	361681 ± 12273	0.9856 ± 0.0017	0.923 ± 0.014	0.827 ± 0.002

near-deterministic model has near-degenerate σ at dropout 0.05, so coverage is delivered by a **near-constant** band: the conformal quantile is enormous ($\approx 1.5\text{--}1.7 \times 10^9$) and the reliability ECE is ≈ 0.31 . Replacing it with a deep ensemble of $M=5$ independently trained Transolvers—per-cell standard deviation across members as σ —converts that into a genuinely input-adaptive band while preserving coverage ([Table 8](#)): on the same 100/100 split the quantile drops to $q = 2.35$, ECE improves to **0.074**, and coverage holds at 0.915 (consistent with the $\approx \pm 0.03$ binomial spread at $n_{\text{test}} = 100$). This is a single ensemble training run; the calibration/test *split* is not the fragile part—re-drawing it 20 times leaves coverage at 0.899 ± 0.009 with $q = 2.26\pm 0.05$ on pressure ([subsection 4.7](#)). The ensemble mean is also more accurate than any single member ($\rho_{C_l} = 0.9995$, $\rho_{C_d} = 0.998$, C_l/C_d relative force error 4.4%/5.7%) and more accurate than the DEQ-corrected single backbone (`mse_u/v/p` = 0.077/0.058/440 against 0.116/0.079/471), so we do not claim the corrector is the best available accuracy mechanism. The two answer different questions on a cost axis: the ensemble is the better *accuracy* play when $5\times$ the training is affordable, the corrector the better *cheap* play; an M -subset study shows $M=2$ already supports the triage decisions while full band adaptivity needs $M\approx 4\text{--}5$ ([section 5](#)).

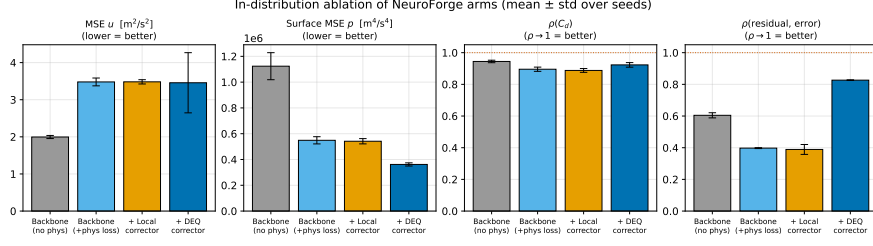


Figure 1: In-distribution ablation (Table 9): the DEQ corrector lifts the trust signal (0.40 → 0.83) and surface fidelity while regressing volume `mse_v/mse_p`.

4.5. The grid backbone: the corrector trades accuracy for trust signal

This ablation uses the **weak grid backbone**; it establishes the backbone-dependence we report as a finding, and should be read against subsection 4.4, where the same corrector *does* reduce field MSE. Here the DEQ corrector is **flat** on volume `mse_u` (3.457 vs 3.479; per-seed deltas sign-inconsistent, 2/3, so noise) and **worse** on `mse_v` (+129%, 3/3, non-overlapping distributions) and `mse_p` (+57%, 3/3); the feed-forward `local` corrector helps on **nothing**. The DEQ arm buys surface-pressure MSE (−34%, 3/3) and a slightly better drag ranking (0.895 → 0.923, 3/3) at a measured cost to volume velocity and pressure. We do **not** claim the corrector improves accuracy in aggregate on this backbone.

The detector succeeds, and a control sharpens the thesis. The same DEQ arm roughly **doubles** the residual-error Spearman correlation, 0.397 → 0.827 (3/3, std ≈ 0.002, Cohen’s $d \approx 175$; effect sizes use the population std, matching the tables)—the largest, cleanest effect in the table. Separately, removing the physics loss entirely **improves** `mse_u`, `mse_v`, `mse_p`, ρ_{C_l} , ρ_{C_d} and the residual-error correlation (all 3/3), at the cost of **doubling** surface-pressure MSE (549k → 1124k, 3/3). The physics-loss-free backbone is in fact the **best force-ranking model in the table** (ρ_{C_d} 0.945, ρ_{C_l} 0.992)—better than any corrected model, with no correction loop at all—so we explicitly do **not** claim the loop delivers best-in-class force ranking. The physics objective buys surface-pressure fidelity and a strong trust signal, not volume accuracy or force ranking on this weak backbone.

Version note for readers of the preprint. The single-seed figures reported in earlier public versions— ρ_{C_l} rising 0.924 → 0.958 and surface MSE −25% under the corrector—do not replicate: across 3 seeds the backbone already sits at $\rho_{C_l} = 0.987$ and the DEQ corrector at 0.986, flat-to-slightly-worse and sign-inconsistent. Table 9 supersedes them.

Table 10: Out-of-distribution AirFRANS ablation (regime-disjoint train→test), 3 seeds, mean $\pm$ std. The full four-arm table is in [results/full_research/ood/ablation_ood.md](#). See [Figure 2](#).

task	arm	mse_u	mse_v	mse_p	surf. mse_p	ρ_{C_t}	ρ_{C_d}	resid↔err ρ
reynolds	backbone	16.218 $\pm$ 0.746	1.187 $\pm$ 0.149	7220.7 $\pm$ 612.2	964641 $\pm$ 61419	0.950 $\pm$ 0.008	0.893 $\pm$ 0.009	0.748 $\pm$ 0.077
reynolds	+ DEQ	5.300 $\pm$ 1.544	1.208 $\pm$ 0.255	6469.5 $\pm$ 438.0	652281 $\pm$ 45390	0.943 $\pm$ 0.008	0.915 $\pm$ 0.009	0.742 $\pm$ 0.041
aoa	backbone	4.312 $\pm$ 0.077	1.310 $\pm$ 0.039	6042.8 $\pm$ 242.5	2000592 $\pm$ 55092	0.963 $\pm$ 0.003	0.926 $\pm$ 0.002	0.314 $\pm$ 0.064
aoa	+ DEQ	3.538 $\pm$ 0.681	1.407 $\pm$ 0.089	4620.6 $\pm$ 170.2	1042711 $\pm$ 78298	0.966 $\pm$ 0.006	0.905 $\pm$ 0.013	0.746 $\pm$ 0.027

4.6. Residuals detect under distribution shift

We evaluate each arm on the regime-disjoint **reynolds** and **aoa** splits (trained on the train range, tested on the held-out range—true extrapolation); [Table 10](#) reports both.

The trust signal is regime-invariant. On the **aoa** split the bare backbone’s residual–error correlation **collapses to** 0.314, while the DEQ-corrected model holds it at 0.746 (3/3, Cohen’s $d \approx 8.9$, non-overlapping: every corrected seed ≥ 0.72 , every backbone seed ≤ 0.38). On **reynolds** both are high (≈ 0.74 – 0.75). The residual is a reliable detector exactly where detection matters most. The same ablation shows the DEQ corrector reducing the in-distribution→OOD *gap* on volume velocity, volume pressure and surface pressure (e.g. **reynolds** mse_u 16.2 $\rightarrow$ 5.3, **aoa** surf. mse_p 2.00M $\rightarrow$ 1.04M, both 3/3), which we report as an observation about the loop’s behaviour and **not** as a generalisation guarantee, preserving two negatives. (i) **mse_v is an attenuated-regression artifact, not a gain**: the corrector’s *absolute* OOD mse_v is worse than the backbone in every regime (0.880/1.208/1.407 vs 0.385/1.187/1.310); its gap only looks smaller because its in-distribution value is already inflated. (ii) **Drag ranking is task-dependent**: ρ_{C_d} improves on **reynolds** (0.893 $\rightarrow$ 0.915, 3/3) but **regresses on aoa** (0.926 $\rightarrow$ 0.905, worse on all 3 seeds). Methodological caveat: the in-distribution reference is the **full** split, so the gap mixes a train-set change with regime shift, and the **aoa** mse_u effect, while 3/3 directional, is seed-0-dominated.

A second dataset: the trust signal generalises beyond AirFRANS. We evaluate the residual trust signal on **DeepCFD** [43]—2-D laminar incompressible flow over bluff bodies: a different geometry family *and* a different regime. We train an FNO backbone (3 seeds) on an out-of-distribution split holding out the largest bodies and measure the same per-case residual–error Spearman correlation ([results/deepcfde/deepcfde_results.json](#)). The trust signal holds ([Figure 3](#)): 0.770 ± 0.119 , comparable to—indeed above—the AirFRANS Transolver value (0.611). The error spread is genuine (per-case rel- L_2 median 0.063, max 0.136), and per-seed case-level bootstrap 95% CIs

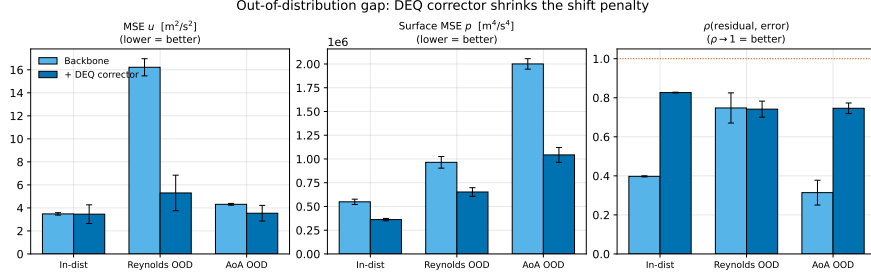


Figure 2: OOD gap and the regime-invariant trust signal ($\text{aoa } 0.31 \rightarrow 0.75$); the bare backbone’s signal collapses, the corrected model’s holds.

(10^4 resamples) are $[0.65, 0.77]$, $[0.58, 0.72]$ and $[0.91, 0.95]$, all far from zero (`results/control/bootstrap_spearman_ci.json`). A mechanistic reason it can be *sharper* than the turbulent case: laminar flow has $\nu_t = 0$, so the residual is the *exact* Navier–Stokes residual with no closure ambiguity.

A geometry the model never saw (qualitative). We run the airfoil-trained Transolver on a *cylinder* and ask not whether the prediction is good (it is not) but whether the residual registers the geometry as out-of-distribution. It does, in the residual *magnitude*: normalising each case’s residual by its freestream U^2 , the dimensionless far-field residual is $\approx 7\times$ higher on the cylinder (median $\approx 30\times$; [Figure A.8](#)). The gap holds in the *far field*, away from any wall, so it is a genuine geometry-OOD effect rather than a near-wall discretisation or velocity-scale artifact. We report a control that *refutes* the tempting spatial reading: the residual’s bright near-body ring is a generic no-slip-wall feature, not an OOD localiser—in-distribution airfoils with good predictions show the same near-body-vs-far-field concentration ratio (≈ 23) as the cylinder (≈ 18). The out-of-distribution signal is the field-wide residual *magnitude*, not its spatial ratio.

4.7. The conformal trust layer: coverage, contraction, and what the floor costs

Contraction. Iterating the learned DEQ operator from a random initialisation on 24 real AirfRANS cases, the measured per-step ratio $\|\delta_{k+1} - \delta^*\| / \|\delta_k - \delta^*\|$ (in the geometric regime, before the $\sim 10^{-7}$ solve floor) has **median 0.78** and maximum 0.86—strictly below the design bound $\kappa = 0.9$ of [Proposition 2](#)—and reaches a relative distance of 10^{-5} to the fixed point in a median of 37 steps (`results/certificates/h5_contraction.json`). The corrector contracts as designed; this is a property of the *operator*, separate from whether minimising the output’s PDE residual helps the field (it does not, [subsection 4.10](#)).

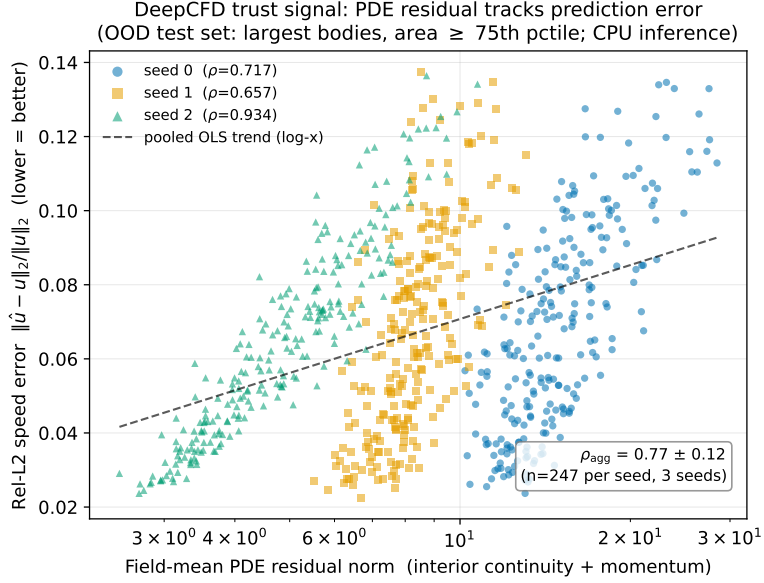


Figure 3: The trust signal generalises to a second dataset (DeepCFD, 2-D laminar flow over bluff bodies; OOD held-out-largest-body split, 3 seeds). Per-case field-mean PDE residual norm vs. rel- L_2 speed error: a low residual tracks a low error, as on AirFRANS. Per-case Spearman $\rho = 0.77 \pm 0.12$ (per seed 0.72/0.66/0.93; the trend holds *within* each seed).

In-distribution coverage. With per-cell MC-dropout σ (16 passes) and a 100/100 calibration/test split, split-conformal calibration at $\alpha = 0.1$ attains per-channel coverage of **0.911** (u), **0.928** (v), **0.942** (p)—all in the $[0.85, 0.95]$ band and conservatively above the 0.90 target, as the $\geq 1 - \alpha$ guarantee requires. The conformal multipliers ($q = 0.77, 1.30, 1.75$) both tighten an over-dispersed and widen an under-dispersed raw σ , so the calibration is doing real work, and coverage tracks $1 - \alpha$ across a sweep of α (Figure 4). The reliability diagram is exact at the target level but over-covers at lower nominal levels (ECE $u/v/p = 0.064/0.093/0.130$), the signature of a heavy-tailed $|\text{error}|/\sigma$ ratio; we therefore claim coverage at the chosen α , **not** full distributional calibration.

The certificate is calibrated on the deployed, corrected field. Split-conformal calibration applies to whatever predictor it scores, and a raw-backbone certificate does **not** transfer unchanged through a corrector. A within-run frozen- q contrast on the dropout-FNO—same q , same input-conditional σ , same cases, only the raw vs. corrected error map (15/15 split)—shows the direction of the coverage change tracking the direction of the error change: where the corrector enlarges the error the raw band drops below target (v

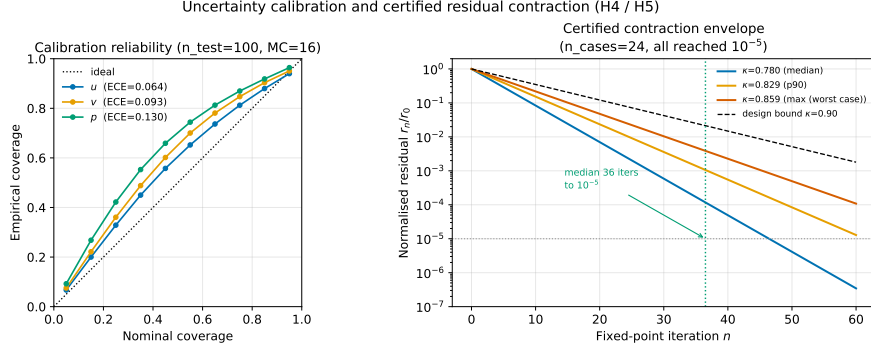


Figure 4: In-distribution reliability / coverage-vs- α and the measured DEQ contraction ($0.78 < 1$).

0.896 $\rightarrow$ 0.819, p 0.864 $\rightarrow$ 0.777), and where it shrinks the error the band over-covers (u 0.856 $\rightarrow$ 0.888). Re-fitting q on the corrected calibration field restores coverage (v 0.819 $\rightarrow$ 0.894, p 0.777 $\rightarrow$ 0.834) and sharpens it where it was loose (u 0.888 $\rightarrow$ 0.874). These estimates are directional at $n = 15$ ($\approx \pm 0.05$ standard error) but reproduce on an independent 20/20 split. On the deployed Transolver we verify it *directly*: with σ the frozen five-member ensemble std and the scored field replaced by the DEQ-corrected output, at the production 100/100 split and across all three corrector seeds (`results/uq_ensemble/w2_conformal_corrected.json`; a faithfulness gate reproduces the published backbone certificate, $q=2.35$, coverage 0.915, exactly), re-fitting q gives a valid certificate (coverage $u/v/p = 0.88/0.91/0.91$ against the 0.90 target), and because the corrected field’s error is lower at frozen σ that accuracy propagates into proportionally tighter intervals (q : v 2.58 $\rightarrow$ 2.34, p 2.35 $\rightarrow$ 2.10 on every seed). Applying the *backbone’s* own q to the corrected field over-covers on v and p (≈ 0.93) while u holds at ≈ 0.88 , inherited from the backbone’s own heavy-tailed u band (0.881) rather than introduced by the corrector. Re-drawing the split 20 times (`results/uq_ensemble/multisplit_conformal.json`) leaves coverage tightly centred on target in *every* arm and channel (0.895–0.902, split-to-split std ≤ 0.014); the corrected-field sharpening is robust, with pressure q dropping from 2.26 ± 0.05 to 2.02–2.07 and v from 2.55 ± 0.05 to 2.28–2.31, while the heavy-tailed u channel is mixed.

Out-of-distribution coverage (empirical, not a guarantee). Calibrating q on the in-distribution full split and evaluating on the OOD splits, coverage stays in the target band: u 0.90/0.91/0.91, v 0.91/0.94/0.92, p 0.92/0.94/0.90 (full / reynolds / aoa) (Figure 5). **This is an empirical observation, not a guarantee.** The conformal guarantee assumes exchangeable calibration

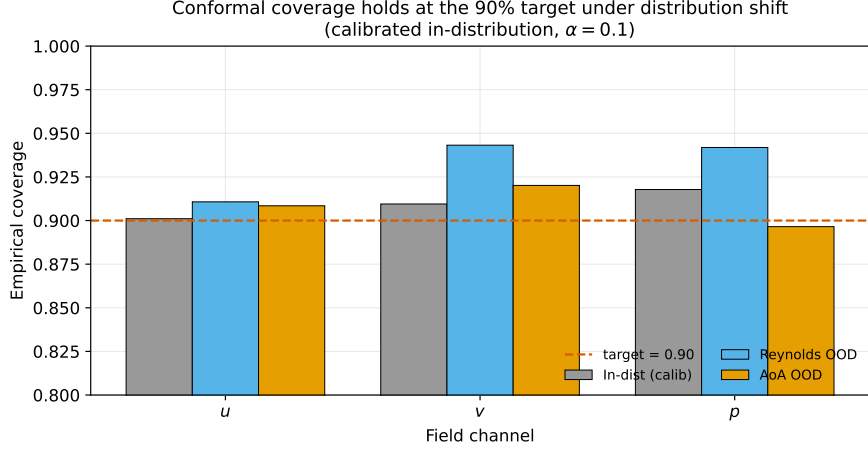


Figure 5: Conformal coverage stays in-band on OOD splits (empirical, via σ inflation).

and test draws, which fails under shift; coverage nonetheless holds because the MC-dropout σ inflates appropriately on shifted inputs, so the same q still brackets the larger errors.

What the floor costs the certificate: valid, but wide. The quantity an engineer actually wants bounded is a force coefficient. Conformalising the monitored residual against $|\Delta C_D|$ over 400 random half-splits at a 0.90 target on the deployed backbone gives coverage 0.901–0.903 across three seeds—so the guarantee holds, and the floor does *not* invalidate split conformal, which needs only exchangeability. What the floor destroys is tightness: the median bound is **6.5–7.9 $\times$** the median drag error it certifies. We tested the obvious explanation—that the floor is what makes it wide—and it is *wrong*: removing the floor exactly, by scoring $\|R_h(\hat{u}) - R_h(u^*)\|$, makes the bound **wider** (7.35 $\times$ to 11.33 $\times$, 3/3 seeds; a scale-invariance gate passes at exactly 0.0). What sets the width is the heavy tail of the drag error itself: its 0.9 quantile is 15.1 $\times$ its median, so an uninformative score would give 15.1 $\times$. Against that reference the monitor is worth **2.34 $\times$** , and that is the honest statement of what it buys.

4.8. From correlation to decisions, and where the physics earns its place

A rank correlation of 0.6 can sound modest, so we measure what the trust signal is *for*: deciding which predictions to reject (results/selective/selective_prediction.json; Figure 6). On the 200 AirfRANS test cases (ensemble-mean arm) the residual detects worst-decile-error cases with AUROC 0.871; the ensemble σ alone gives 0.894; the rank-fused score reaches **AUROC** 0.905 (AUPRC 0.681 against a 0.10 prevalence

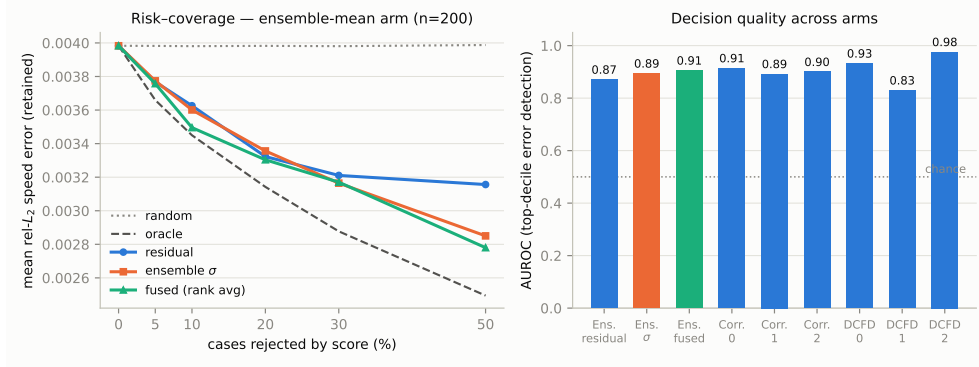


Figure 6: **Selective prediction with the trust signal.** *Left:* risk-coverage curves on the AirfRANS test set (ensemble-mean arm): mean retained relative- L_2 error after rejecting the least-trusted fraction of cases by physics residual, ensemble σ , or their rank fusion, against the oracle (reject by true error) and random rejection. *Right:* AUROC for detecting worst-decile-error cases across arms (AirfRANS ensemble-mean, three corrected seeds, three DeepCFD OOD seeds); chance = 0.5.

baseline). The same residual score transfers: AUROC 0.89–0.91 on the three DEQ-corrected arms and 0.83–0.98 on the three DeepCFD OOD seeds. In risk-coverage terms, rejecting the 10% least-trusted cases by the fused score reduces the retained mean relative- L_2 error from 0.0040 to 0.0035—recovering $\approx 91\%$ of the error reduction an *oracle* rejecting by true error could achieve at that budget (the residual alone recovers $\approx 67\%$; random rejection recovers none). Case-level bootstrap 95% CIs (10^4 resamples) put the ensemble-arm residual correlation at 0.610 [0.506, 0.698] and the fused score at 0.703 [0.615, 0.773]; every arm’s CI excludes zero by a wide margin.

On field error the physics does not earn its place; on drag it does. A *physics-free* score—the ensemble σ , which needs no operator, no mesh and no PDE—is nominally ahead of the residual (0.894 against 0.871), and a paired case-level bootstrap cannot separate them ($\Delta\text{AUROC} +0.023$, 95% CI $[-0.035, +0.083]$). The gain that survives is the *fusion*’s. There is one place the physics wins outright, and it is the place an engineer cares about: ranking cases by error in the *drag coefficient* rather than in the field, the residual reaches AUROC **0.952**, beating the physics-free score decisively ($\Delta\text{AUROC} +0.055$ to $+0.088$, bootstrap CIs excluding zero).

The detector is not a difficulty proxy. Cases whose geometry rasterises badly could be both high- $\|r^*\|$ and hard to predict, which would make the correlation a restatement of case difficulty. Conditioning on each case’s *own* floor—the ground-truth field’s residual, an oracle quantity—leaves a partial rank correlation of 0.561 against a raw 0.610, so 92% of the association survives

and the drop’s CI includes zero. The deployable residual also significantly outranks that oracle difficulty variable itself, which a pure difficulty proxy could not do.

The acceptance gate is a throttle, not a filter—and what it buys is the certificate. Over 1200 gated steps (200 AirfRANS **full** test cases $\times$ 3 seeds, on both the deployed Transolver+DEQ system and the ensemble-mean path, with the accept rule, backtrack count and tolerance imported from the shipped `correction_loop; results/control/acceptance_gate.json`), the test admits a step on 99.8% of deployed cases—the *full* correction on 50.7% of them—and refuses outright on one case in 600. The mechanism is backtracking: where the full step would raise the residual it is refused, but a halved step lowers it and is admitted (on the ensemble path the residual ratio falls from a median 1.027 at the full step to 0.975 at half). The admitted step lowers true rel- L_2 error on 89.3% of deployed cases, by a median **5.8%** (74.3% and 2.6% on the ensemble-mean path, where the correction has less headroom). But the obvious control—an *ungated* fixed half step, never consulting the residual—improves true error on 95.8% of deployed cases by a median 6.2%. **The gate’s accuracy advantage is therefore not there;** what improves the field is damping the correction, and the residual test is not what earns that. What the gate uniquely buys is the **certificate**: the ungated half step raises the monitored residual on 1.0% of deployed cases and 8.8% on the ensemble path (`results/review/control3_fixed_step.json`), and the gate makes that outcome impossible by construction rather than improbable in practice. On the path a user actually runs the guarantee binds on roughly one case in a hundred: it is a guarantee, not an accuracy claim, and that is what we claim for it. (Scope: a single gated step applied counterfactually to the deployed DEQ correction, which the shipped DEQ path bypasses by design; not a re-run of the multi-iteration feed-forward loop of [Table 9](#), whose corrector checkpoints were not retained.)

4.9. Computational cost

A trust layer is only deployable if auditing costs materially less than the prediction it audits. We time every stage of the deployed pipeline on the AirfRANS **full** test set at resolution 128 (`results/control/inference_cost.json`; 100 cases, median of two timed repeats after an untimed warm-up, CUDA synchronised around each repeat, on one RTX 4070 Ti with the package’s single-thread BLAS cap in force). [Table 11](#) reports the result.

The audit costs 1.13ms against a 3822ms solve—one part in 3400, or **0.03%**; a conservative independent 200-case re-measurement under CPU con-

Table 11: **Per-case wall-clock cost of the deployed pipeline** (128² grid, 100 AirfRANS full test cases, medians). *Where* is the device each stage actually runs on. The audit is the entire certificate: residual maps, trust map, case-level score and conformal band. The classical anchor is the cost of generating one AirfRANS case, as reported by its authors (≈ 25 min on 16 CPU cores); it is quoted for scale, not as a matched-hardware benchmark.

stage	where	median	p95	share of a deployed solve
encode (geometry $\rightarrow$ input planes)	CPU, 1 thread	1413ms	1620ms	preprocessing
backbone forward + rasterise	GPU	2080ms	3537ms	54.4%
correction (Neural Residual Iteration)	GPU	1747ms	2950ms	45.7%
deployed solve (backbone + correction)	GPU	3822 ms	6400ms	100%
audit (residuals + trust + score + band)	CPU, 1 thread	1.13 ms	2.49ms	0.03%
deep ensemble, $M=5$ (for the fused score)	GPU	10 114ms	10 546ms	265%
classical steady-RANS solve [1]	16 CPU cores	$\sim 1\,500$ s	—	$\sim 39\,000\%$

tention gives 1.65ms (`results/control/audit_cost.json`). End to end a case costs 5.24s against the ~ 1500 s that produced its label. The deep-ensemble spread does not share that property: measured by loading five independent backbones, $M=5$ costs $4.86\times$ a single backbone—linear in M , confirmed rather than assumed—which is 2.1 deployed solves of *extra* compute. This sharpens subsection 4.8 into a deployment choice: the residual alone recovers $\approx 67\%$ of the oracle’s achievable error reduction at essentially zero marginal cost, while fusing it with the ensemble spread reaches $\approx 91\%$ but triples the compute budget. The audit’s operation count is $O(N)$ in the number of grid cells; empirically, over a $16\times$ range in N (64^2 to 256^2) it rises only $6.3\times$ (log–log slope 0.64, $R^2 = 0.99$), sub-linear because per-call fixed costs still dominate at these sizes, so its share *falls* as grids grow. These GPU figures are conservative upper bounds rather than a tuned inference benchmark—the backbone stage times the Transolver forward pass *and* a single-threaded NumPy rasterisation, and the encode stage is unoptimised preprocessing that any real design loop would cache. The classical anchor is from the AirfRANS authors on 16 CPU cores, so the $\sim 286\times$ end-to-end figure indicates the order of the gap, not a controlled speed-up. None of these caveats touches the ratio the paper depends on—audit against solve—because both sides were measured in the same run, on the same cases, under the same thread caps.

4.10. What the physics residual can and cannot do

The obvious response to subsection 4.1 and subsection 4.3 is that the metrics should be fixed with physics: score a surrogate by how well its field satisfies the governing equations, and the metadata null loses its purchase. We ran that experiment and it does not work. This section reports the

measured answer in the compressed form; the theorem behind it and the full control set are deferred to a companion paper.

The ground truth fails the check. On 200/200 real AirfRANS test cases the monitored residual of the *exact* field is substantially nonzero ($\|r^*\|$ mean **0.192**, median 0.133; `results/certificates/residual_floor_realdata.json`), while the physically wrong uniform freestream scores an **exact zero** at every case and every resolution, because a spatially constant field has identically zero finite differences. The objective therefore strictly prefers a wrong field to the truth. This is not an artifact of resolution: rasterising the same truth at 128^2 , 256^2 and 512^2 and scoring a region fixed in *physical* units—a reading registered before the run—the floor *rises*, $0.0624 \rightarrow 0.0779 \rightarrow 0.1067$ while h falls $4\times$, in **21/24** cases at a fitted exponent $p = -0.387$, with a second independent 16-case five-rung implementation finding it decaying in 0 of 16. Four controls survive, each with its own scope. Method of manufactured solutions on the same monitor gives $p = 2.06$, so the discretisation converges as claimed—but the manufactured field is a potential flow, on which the monitored viscous term vanishes identically, so this control bounds truncation error and is *not* evidence that the operator is consistent with the one that generated the labels. Restoring the omitted closure term $(\partial_j \nu_t)(\partial_j u_i + \partial_i u_j)$ moves each rung’s mean floor by -0.01% to $+0.18\%$, and the term is only 3.5–4.8% of the floor. A cubic rather than linear interpolant *raises* the floor by 6–12%, though that control bears on the second-derivative block—5–7% of the floor—and not on the first-derivative block carrying 93–95% of it. And a manufactured analytic solution pushed through the identical rasteriser reproduces the effect $\sim 280\times$ smaller, though its fine end reverses too, so it does not show the pipeline cannot manufacture a rising residual. We name two contributions to the measured size—the mismatch between our Cartesian stencil and the body-fitted discrete balance AirfRANS actually solved, and the reconstruction of that reference onto our grid (at fixed h , thinning the source cloud eightfold raises the floor in 16/16 cases)—and separate neither.

So the residual cannot be descended. Minimising $J = \frac{1}{2}\|R_h(y)\|^2$ directly, with no network in the loop, from the *exact ground truth*—so the standard reply that a surrogate is over-smoothed has nothing to attach to—drives the field error up on **200/200** cases under Armijo-line-searched gradient descent (500 steps, $n = 200$, far-field and near-wall Dirichlet data pinned at truth, so J is monotone non-increasing by construction). J falls to 39.8% of J_0 while the error never returns to its starting value at any step, ending at

$1.57\times$ the *entire* error of the deployed Transolver (`mse_u` $0.000 \rightarrow 0.499$; $\|R_h\|$ $0.191 \rightarrow 0.125$). Started from the deployed Transolver the residual falls 34% while the error rises on **96.5/94.0/98.0%** of cases across three seeds (median $+76.5/+73.9/+86.9\%$; Wilcoxon $p < 10^{-30}$), against the supervised corrector’s $-8/-21/-25\%$ on the same backbone. The verdict is unchanged under three step rules—Armijo, fixed η swept over 12 decades, and Adam—and is monotone in the *achieved* J : the harder the objective is minimised, the worse the field. Unpinning the boundary data makes the negative stronger (100/99.5/100%), so the pinned arm is the conservative one.

What holds universally is iterate selection, and where the exception lies. We do *not* claim residual descent always harms the field: from a perturbed start it *reduces* error in 18/24 cases before overshooting, and on the far-from-converged dropout-FNO it lowers error on 75% of cases. What holds universally is a statement about **iterate selection**: the error-minimising and residual-minimising iterates differ in 24/24 cases on both arms, and the residual-selected iterate is strictly worse in **24/24**. The exception is measurable in $\rho = \|R_h(\hat{u})\|/\|r^*\|$ —over 200 cases the fraction on which descent helps rises monotonically with ρ , and the deployed backbone sits at median $\rho = 1.13$ – 1.14 with 88% of cases in the lowest bin, already inside the regime where descent almost never helps. Even there the objective is unusable: the gain is $\approx 1/35$ of the supervised corrector’s, and there is **no stopping rule**, since J falls monotonically on 100% of cases while the error minimum is interior on 78.5%.

The boundary is the operator, not the problem class. Residual-driven correction is not doomed in general. Lei et al. [8] take a surrogate prediction as an initial guess for Newton–Krylov correction on *steady* CFD and lower the residual by over seven orders of magnitude *while* reducing field and aerodynamic error; Huang and Perdikaris [16] report the same on transient rollouts. In those works the residual being driven is the *solver’s own* discrete operator, whose root the reference solution is, so its floor at the truth is zero by construction. Ours is the affordable surrogate-side one, and its minimiser is displaced from u^* . The obvious alternative boundaries—steady versus transient, resolved versus under-resolved—are falsified by Lei et al. [8] being steady and successful. That a PDE residual need not vanish at the ground truth is not ours alone: Zhang et al. [44] report it for shock physics and conclude a physics-informed loss “must be defined relative to the residual present in the ground-truth data”—a remedy requiring $R_h(u^*)$, which a deployment-time monitor does not have. What the residual *can* do it does

well and cheaply: it ranks (subsection 4.8) and it certifies at a priced width (subsection 4.7).

4.11. Reading against the pre-registered hypotheses

H1 (the corrector improves accuracy): *backbone-dependent*—not supported as pre-registered on the weak grid backbone, supported on the SOTA Transolver (all three volume channels down on all 5 seeds, Table 7). **H2 (the residual is a valid trust signal):** *supported, robustly*, in and out of distribution ($\rho > 0$ for every arm/seed/split; DEQ lifts it to 0.83 in-distribution and rescues the aoa collapse $0.31 \rightarrow 0.75$), with two qualifiers stated in full in subsection 4.8. **H3 (DEQ $\geq$ feed-forward corrector):** *holds on the design axis only*—DEQ wins on surface MSE, ρ_{C_d} and the trust signal (3/3) and loses on volume `mse_v/mse_p`. **H4 (conformal coverage):** *supported in-distribution, empirical only* OOD; coverage was never in doubt, width is what the floor costs. **H5 (DEQ contraction):** *supported* (measured factor $0.78 < \kappa = 0.9 < 1$). **Not pre-registered—the residual as a correction objective:** the registered set predates that experiment; we record for completeness that had it been registered it would be *falsified* (subsection 4.10). Every number comes from the committed, reproducible harness (`benchmarks/ablation.py`, `scripts/run_certificates.py`, `results/sensitivity/`).

5. Limitations

- **Every field number in subsection 4.1 is scored on the r_{128} raster, and that representation cannot adjudicate the boundary layer for either arm.** The rasterised ground truth, resampled at the native cloud nodes, commits a round-trip error $413\times$ larger than Transolver’s own per-node error pooled over the scored crop, and $495\times$ larger inside $0.005c$, where 43% of the dataset’s nodes live. Two things follow. First, the *weighting* of that raster is not a neutral choice, and we therefore report both weightings rather than one (Table 3); the ranking reverses between them. Second, no re-weighting reaches the representation: a method right at sub-grid scales earns no credit under either. A frozen $128^2/256^2/512^2$ ladder settles that the *localisation* is not an artifact of the deployed grid—the domain fraction reproduced without learning, the R^2 floor beyond $0.05c$ and the absolute outer-region MSE are flat across a $16\times$ increase in cell count, and at 512^2 the error is *more* tightly localised—while the near-wall *share* eases from 0.92/0.90 to 0.88/0.84 and misses a pre-declared 0.85 floor on v at the finest rung, so the registered verdict is PARTIAL and we report it as PARTIAL. What

remains open is a head-to-head at native *resolution*. The deployed checkpoints do store both normalisers (`point_norm` and `grid_norm` in `checkpoints/v2_transolver/seed{m}.pt`) and we load them, so the obstacle is not a missing normaliser; it is that scoring the interpolator at native nodes changes the measure and the representation at once, penalising it for sampling a $0.0234c$ grid at wall nodes—a handicap unrelated to either question above. That experiment is the single most valuable follow-up in the paper and is not reported here. Relatedly, [Table 2](#) divides by variances computed on the *r128* raster rather than in point space, so it is faithful in spirit but not byte-identical to the training normaliser; the ratios between rows are unaffected, since the divisor is shared.

- **No matched surrogate row on the extrapolation splits.** The degradations in [subsection 4.2](#) are the interpolator’s own in-distribution-to-OOD changes, so they must not be read as “interpolation generalises better than a surrogate”. Two scope notes: the `reynolds` test split is the first 200 of 496 cases (the *r128* cache limit), an unbiased subsample whose U/α ranges match the full set but a subsample nonetheless; and `aoa` has 196 test cases, not 200.
- **The interpolator is a baseline, not a method.** It has zero parameters but carries 210 MB of training fields resident at inference, and it is deterministic given the split, so the headline rows have no seed variance to report (the train-subset ladder uses five random subsets per size for $n < 800$). We make no deployability claim for it.
- **The covariate null is one benchmark, one metric family.** We compute it for AirfRANS lift and drag only. Extending it to DrivAerNet++, DrivAerML and AhmedML is straightforward—all three publish their design parameters—and is not done here. We also do not refit the null inside the `scarce`, `reynolds` and `aoa` splits, which would be required before comparing published per-split force numbers against it.
- **Backbone-dependent correction quality, and a third backbone that is not like-for-like.** The supervised DEQ corrector reduces field MSE by 8–25% on the SOTA Transolver but is flat-to-unhelpful on the weak grid backbone; we report both and claim only the latter. The trust signal is positive on all three backbones tested ($\rho = 0.611$ Transolver, ≈ 0.40 – 0.83 grid backbone, 0.851 bare MeshGraphNet), but the MeshGraphNet arm was trained on 16k-point subsamples and

evaluated at $\approx 180\text{k}$ and our own density control returns DENSITY-DRIVEN, so the matched-density range is $\rho \approx 0.40\text{--}0.61$. We make no competitiveness claim for the grid backbone; the Transolver backbone we run *on* is itself competitive.

- **Grid resolution, near-wall blindness, and what that costs drag.** A uniform 128^2 Cartesian grid cannot resolve a $Re \approx 10^6$ boundary layer, so wall quantities are approximate and the absolute MSE values are not comparable to body-fitted solvers. The monitored residual additionally zeroes the solid-adjacent ring—where the central stencil reaches into the $u = v = 0$ discontinuity—so the trust signal operates on the bulk and is structurally blind to the immediate near-wall band, precisely where near-wall error concentrates. Consequently the *viscous* component of drag is unavailable on this representation, and every drag statement here concerns the pressure-dominated component.
- **Eddy-viscosity accuracy is backbone-dependent.** Against ν_t 's own fluid-masked variance the channel is well predicted on the grid backbone ($R^2 = 0.996$) but only moderately on MeshGraphNet ($R^2 = 0.67\text{--}0.70$). Because $\nu_t \approx 5.2\nu$ supplies $\approx 84\%$ of ν_{eff} , a backbone with a weak ν_t carries that error straight into the monitored residual—the audit's operator is only as turbulent as the closure the surrogate predicts.
- **The resolution ladder has its own ceiling, and we do not explain the rise.** The floor's growth is measured over $128^2\text{--}512^2$, and at the finest level the raster spacing is only $1.27\times$ the source cloud's far-field nearest-neighbour spacing, so a still finer level would begin differentiating the interpolant rather than sampling the solution. The $128^2 \rightarrow 256^2$ leg is clear of that limit and already shows the rise, but we claim no asymptotic rate—only that the floor does not decay over the range a 128^2 -trained surrogate is deployed in. We also stop short of naming the mechanism: holding h fixed and thinning the source cloud eight-fold raises the floor in 16/16 cases ($d \log \|r^*\| / d \log s = +0.55 \pm 0.27$), which rules out attributing the growth to operator provenance alone, and the registered representation-error model (predicted exponent 1.64, rise $4.8\text{--}8.0\times$) is refuted by the measured $1.8\times$.
- **Scope of the descent result.** The $n=24$ arms use Adam with per-channel scaling, so the field error rises overall but not monotonically and those claims are about endpoints rather than paths; the $n=200$ Armijo arm closes this. The headline arm also freezes ν_t : descending the eddy

viscosity alongside (u, v, p) quarters the velocity damage, but only by dumping the residual into the closure—on the deployed backbone (seed 0) `mse_nut` rises seven orders of magnitude ($6.3 \times 10^{-9} \rightarrow 0.115$) while the field error still worsens on 82.5% of cases—so we freeze it in the headline and report the four-channel arm as a control. The full evidence, including the resolution ladder, the four artifact controls and the theorem, is deferred to a companion paper.

- **The residual fails as an objective; the gate does not, and neither is the source of the corrector’s gain.** The DEQ contraction guarantee is in the correction variable δ , not in the output’s PDE residual. The acceptance gate is a different mechanism: measured, it admits a step on 99.8% of deployed cases and that step lowers true error on 89.3% of them—though that measurement is a single gated step on the deployed corrector, not the multi-iteration feed-forward loop, which remains untested at this granularity, and an ungated half step does better on accuracy alone. Nor is the 8–25% field-MSE gain attributable to the residual as a corrector *input*: zeroing that input at train and inference (5 seeds) matches or slightly exceeds the deployed corrector (WITH beats NULL on 0/5 seeds), with one caveat stated in [section 3](#)—those channels are built from the wider differentiable stencil, whose larger kernel we have not excluded.
- **Two 2-D datasets; not yet 3-D.** The trust signal is validated on turbulent-RANS AirfRANS airfoils and laminar DeepCFD bluff bodies ($\rho = 0.77 \pm 0.12$), so the phenomenon is not AirfRANS-specific, but all results remain 2-D. Extending the residual verifier to 3-D [\[6, 7\]](#) requires 3-D residual operators and multi-TB, often unsteady datasets beyond a single-GPU budget.
- **Conformal coverage out-of-distribution is empirical, not guaranteed, and adaptivity needs an ensemble.** The distribution-free guarantee assumes exchangeability, which fails under shift; OOD coverage holds in our experiments only because the uncertainty inflates appropriately. Scores are pooled over spatially correlated cells, so the effective sample size is below the raw cell count, and the calibration set is small ($n = 100$). On the near-deterministic Transolver, MC-dropout σ is near-degenerate, so although coverage holds (0.902) the quantile is $\approx 10^9$ and ECE ≈ 0.31 : coverage there is delivered by a near-constant band. A deep ensemble of $M=5$ recovers an adaptive band ($q = 2.35$, ECE 0.074) at $5\times$ the training cost. Neither the split nor M is the remaining caveat—20 re-draws give coverage 0.899 ± 0.009 ,

and sweeping every member subset of size $M = 2..5$ (26 subsets; `results/uq_ensemble/mstudy.json`) leaves the *decision* layer insensitive (fused AUROC 0.905–0.912, coverage 0.905–0.915 at every M , so even $M=2$ supports triage) while band *adaptivity* is what M buys (q inflates $2.35 \rightarrow 3.74 \rightarrow 9.98$ and ECE degrades $0.074 \rightarrow 0.113 \rightarrow 0.199$ as M drops $5 \rightarrow 3 \rightarrow 2$, with $M=4$ already close to $M=5$). Independently retrained ensembles remain future work.

- **Classical fallback is a stub interface.** The trust-gated fallback fixes the integration seam and reports what would run; the OpenFOAM/SU2 backends are not implemented and bear on no quantitative claim.

6. Conclusion

The field-MSE ranking on AirfRANS is a weighting choice, and the benchmark’s primary metric cannot see what a surrogate actually buys. Interpolating seven scalars read off the case name gives $8.4\times$ lower volume-pressure error than a matched-budget Transolver on the area-uniform raster the field tables use, and no advantage on any channel once the identical predictions are re-weighted by the dataset’s own node measure, which places 64.4% of its mesh inside $0.05c$ of the wall against 1.25% of the raster’s area. No published AirfRANS comparison states which weighting it uses. Nor can the raster settle the matter by being refined in place: resampled at the native nodes its own round-trip error exceeds the surrogate’s by $413\times$, so the deployed protocol measures far-field skill for both arms. What the measurement does resolve, on both arms for the first time, is where each method’s error lives—the band ratio on u runs $906\times$ to $0.31\times$, monotone over seven bands, with 71.9% of Transolver’s own squared error beyond $0.5c$ against the interpolator’s 3.6%, and the interpolator reproducing 98.7% of the domain at $R^2 \geq 0.9996$ from seven numbers in a file name. Learning earns its keep in the first cell off the wall and in the closure, which is precisely where the benchmark’s own authors located their models’ failure without quantifying it. Meanwhile a regression on the case name outranks every published drag entry, three of which span zero, and outranks all four published lift entries as well—though on lift the separation is 3.8 and 1.9 of each model’s own published seed standard deviations for two of them and only 1.6 and 0.8 for the other two. We do not read this as evidence that those results are wrong; we read it as evidence that both a field metric and a force-rank metric on this benchmark are uninterpretable without their non-learned reference point—and, for the field metric, without the weighting it is computed under.

Fixing the metric with physics does not work, and the reason is measurable. On 200/200 AirfRANS cases the monitored residual of the exact field is substantially nonzero (mean 0.192) while the physically wrong uniform freestream scores an exact zero, and refining the grid makes the floor *larger*. Descending that residual from the exact ground truth, with no network in the loop, drives the field error up on **200/200** cases. The boundary is the operator, not the problem class: Newton–Krylov corrections succeed on steady CFD precisely because the residual they drive is the solver’s own, whose floor at the truth is zero by construction. What the residual *can* do it does cheaply—the whole certificate costs 0.03% of the solve it vouches for, and triage on the fused score recovers $\approx 91\%$ of an oracle’s achievable error reduction—but on field error a physics-free ensemble score matches it, and the conformal bound it supports is valid and $6.5\text{--}7.9\times$ wide against the $15.1\times$ an uninformative score would give.

The durable contribution is a pair of reference points, a measurement of how much the field ranking depends on a convention nobody states, and the boundary they draw together; not a system. Anyone reporting a field metric on AirfRANS should report it against parameter interpolation, which takes fifteen minutes of CPU time and localises whatever advantage remains to the first cell off the wall—and should say which cell weighting the number is computed under, because on this benchmark that choice moves a pressure ratio by a factor of 19 and reverses its sign. Anyone reporting a force-rank metric should report it against a regression on the case name, which takes seconds. And anyone building a surrogate-side physics check should measure their operator’s floor on their own ground truth first: if it is not near zero, the detector may still be worth having, but the objective is not.

Appendix A. Published AirfRANS baselines, and two context figures

Table A.12 reproduces the canonical AirfRANS-paper baselines as field context. The two figures that follow it are context rather than evidence: we make no competitiveness claim for the grid backbone (Figure A.7), and the cross-geometry probe (Figure A.8) is explicitly qualitative. Every figure supporting a claim the main text makes stays beside that claim.

Code and data availability

The complete source, training/evaluation harnesses, and per-claim reproduction map (docs/REPRODUCE.md) are released as the open-source package neuroforge-cfd (<https://github.com/ali-kin4/neuroforge-cfd>;

Table A.12: Published AirfRANS full-task baselines from Bonnet et al. [1], as field context only. Volume MSE on standardized fields ($\times 10^{-2}$); surface $\bar{p}$ ($\times 10^{-1}$); force relative errors (%); ρ = Spearman rank correlation vs. *official* force labels. Our rows are *not* comparable (physical-unit MSE, self-integrated forces; subsection 4.4). The near-zero/negative ρ_{C_d} is the benchmark’s known drag-ranking difficulty; subsection 4.3 adds the null that makes this table readable.

method	$\bar{u}_x$	$\bar{p}$	surf. $\bar{p}$	C_L err	C_D err	ρ_{C_l} / ρ_{C_d}
MLP	0.95	0.74	1.13	0.77	4.29	0.913 / -0.117
GraphSAGE	0.83	0.66	0.66	0.52	4.05	0.965 / -0.303
PointNet	3.50	1.15	0.93	0.74	14.64	0.938 / -0.022
Graph U-Net	1.52	0.66	0.39	0.49	10.39	0.967 / -0.138

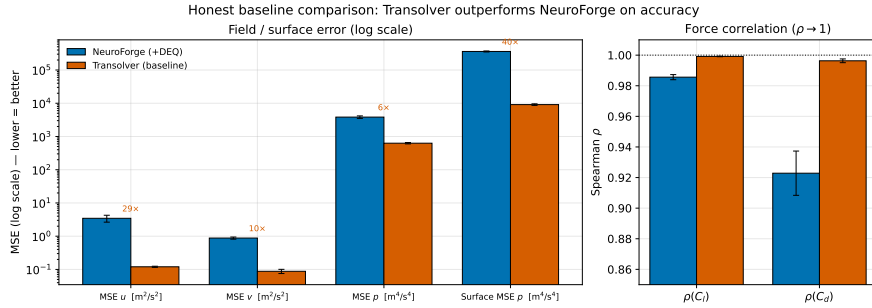


Figure A.7: Matched-budget baseline (Table 1): the grid backbone trails Transolver ~ 4 – $60\times$ across channels (no competitiveness claim).

archived at Zenodo, DOI [10.5281/zenodo.21277928](https://doi.org/10.5281/zenodo.21277928)). Every headline number maps to a committed script and result file, and a top-level manifest (`results/MANIFEST.json`) records seeds, environment, and SHA-256 hashes for hash-drift detection. The two benchmarks are public: AirfRANS [1] and DeepCFD [43]. The package is CPU-first and runs end-to-end with zero downloads via a synthetic data generator; the GPU runs (backbone training) are documented with exact commands and expected outputs.

Use of AI-assisted development tools

Substantial portions of the research software (model implementations, training/evaluation harnesses, and analysis scripts) were developed with the assistance of a generative-AI coding tool (Claude, Anthropic; Claude Opus 4.8 and Claude Fable 5 models, used through the Claude Code environment), operating under the authors’ direction. The authors specified, reviewed, and validated all code; every reported number is produced by a committed script in the public repository and verified against an artifact

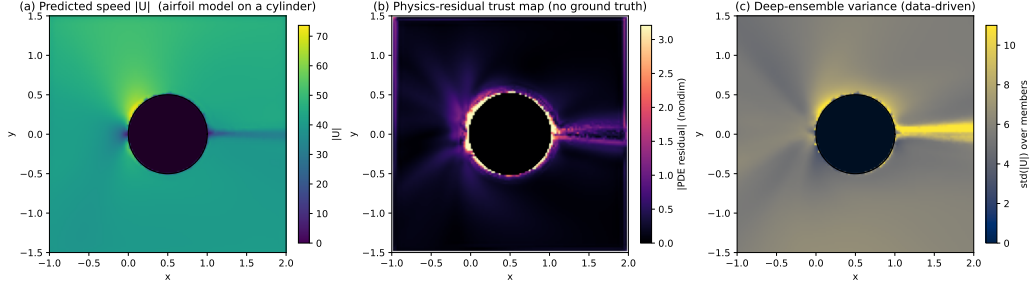


Figure A.8: Cross-geometry OOD (qualitative): an airfoil-trained model on a cylinder. (a) predicted speed (visibly broken); (b) the physics-residual map (no ground truth used), elevated field-wide; its bright near-body ring is a generic no-slip-wall feature common to all bodies, *not* an OOD localiser; (c) deep-ensemble variance. The OOD signal is the field-wide residual *magnitude*— $\approx 7\times$ higher than in-distribution airfoils in the far field, away from any wall (subsection 4.6)—not its spatial concentration.

manifest of SHA-256 hashes (`results/MANIFEST.json`), and the full test suite accompanies the release. The scientific questions, experimental design, pre-registered hypotheses, and interpretation of results are the authors’ own.

CRediT author contribution statement

Ali Jabbary: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Kasra Ghanavati:** Validation, Writing – review & editing.

Declaration of competing interests

The authors declare no competing financial or personal interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist in drafting and editing the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The use of AI assistance in developing the research software is described above.

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